Error Bounds for Logit-based Stochastic User Equilibrium Traffic Assignment

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Abstract

In stochastic user equilibrium (SUE) traffic assignment with fixed path sets and a logit route choice model, we derive bounds on the distance between a given feasible solution and the equilibrium solution. The intent is to provide guidance on termination criteria to reduce run times, which is important because the traffic assignment problem is often a subproblem of a more complex bi-level optimization. Our approach is based on the partial linearization of the SUE objective and provides upper bounds on differences in both aggregate metrics like total system travel time (TSTT) and disaggregate metrics such as link flows and path flows. We demonstrate that the derived path flow, link flow, link cost, and TSTT bounds are tight and cannot be further improved by a fixed factor without additional restrictions on the network topology or problem instance. Results on four city-level networks indicate that, at a relative gap of 10^{-6}, the bounds guarantee that the path flows, link flows, link costs, and TSTT are all within 0.1% of their equilibrium values for a logit dispersion parameter of 0.5, and within 1% for a dispersion parameter of 1.5. Furthermore, we show the error bounds converge to zero at least as fast as the path flows converge in the underlying algorithm used to solve for SUE. We demonstrate the practical utility of the derived bounds in a network design problem to minimize a weighted sum of TSTT and construction cost. Using a local search on the Chicago Sketch network, we show that the bounds allow 96% of the candidate designs to be rejected before their traffic assignment is solved to high accuracy, resulting in a 45% reduction in computation time compared with not using them.

Keywords: Stochastic user equilibrium, error bounds, network design, bi-level programs, traffic assignment

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1 Introduction

Traffic assignment problems (TAPs) predict the routes travelers are likely to choose based on the congestion caused by traffic flow in the network. Algorithms for solving TAPs generally start from a feasible route choice and improve it iteratively toward an equilibrium. Because TAPs are typically formulated as large-scale non-linear optimization problems, an exact equilibrium is reached only in the limit, and finding a highly converged solution is often computationally expensive. Hence, stopping criteria are needed to determine when the current solution is sufficiently close to equilibrium for the iterative procedure to terminate. Gap functions, such as the difference between the average experienced travel time and the shortest path travel time, measure how well the current solution satisfies the equilibrium principle and are therefore widely used as stopping criteria for TAPs. However, these gap functions do not directly quantify how far the current traffic flows are from their equilibrium values. This relationship has been examined empirically in deterministic TAPs by Boyce et al. (2004) and Patil et al. (2021), who investigated how particular gap values relate to the convergence of flows and aggregate measures such as total

system travel time (TSTT). Boyce et al. (2004) suggested that link flows stabilize once the relative gap (RGAP) falls to about 10^{-4}, leading to the widespread adoption of this threshold as a stopping criterion for TAPs. Their analysis, however, considered only one network and one demand scenario. In a more detailed empirical study, Patil et al. (2021) found that aggregate metrics such as TSTT were within 1%–2% of their equilibrium values at an RGAP of around 10^{-4}, while disaggregate metrics required smaller gaps to stabilize. Specifically, they found that 99% of link flows came within 1% of their equilibrium values only at an RGAP of around 10^{-5}, while the entropy-maximizing (most likely) path flows did not stabilize until the RGAP reached approximately 10^{-6}.

The goal of this article is to derive mathematical bounds that complement the existing numerical evidence. In particular, we are interested in theoretical bounds on the deviations of any feasible path flow, and the corresponding link flow, link cost, and TSTT, from their equilibrium values in terms of a gap measure. This would allow us to bound the error in these metrics whenever the TAP is terminated early. Rather than relying on empirical rules of thumb, such as treating an RGAP of 10^{-4} as sufficiently small, we can then terminate a TAP once the guaranteed deviation from equilibrium in the metric of interest falls below a chosen tolerance.

Although modern solvers can find near-equilibrium solutions to a single TAP in seconds or minutes, many practical applications require TAPs to be solved repeatedly, often tens of thousands of times. This is particularly true when the TAP is solved as a lower-level problem within a bi-level optimization framework or as part of a Monte Carlo simulation. In such settings, the derived bounds are especially useful for early termination, allowing individual TAP runs to stop once the desired level of accuracy has been reached in the metric of interest. The computational time saved can then be devoted to evaluating the most promising alternatives, potentially improving the outcome of the overall optimization or simulation framework.

In our approach, we use the stochastic user equilibrium (SUE) model with the logit route choice model to solve the TAP. We will henceforth refer to the problem as the stochastic TAP. Although the studies by Boyce et al. (2004) and Patil et al. (2021) consider deterministic user equilibrium (DUE), a key advantage of SUE for our purposes is that its objective is strictly convex in the path flows (Fisk 1980). Consequently, unlike DUE, SUE has a unique equilibrium path flow solution, and a small gap between the current and equilibrium objective values implies that the current path flows are close to equilibrium. Building on this observation, for a given feasible path flow vector $\boldsymbol{h}$, we use $\boldsymbol{h}$ and its corresponding target flow vector $\mathcal{L}(\boldsymbol{h})$ to compute a gap measure $\gamma(\boldsymbol{h})$ that upper bounds the difference between the current and equilibrium SUE objective values. Here, the target flows represent the route choices travelers would make under the logit model given the costs associated with the current flows. We then use the error bound on the SUE objective to bound the distance between $\boldsymbol{h}$ and the equilibrium path flow solution, and use this path flow bound to derive bounds on the deviations of link flows, link costs, and TSTT from their equilibrium values. In summary, we make the following contributions to the literature:

- For any given feasible solution to the stochastic TAP, we derive upper bounds on the errors in the SUE objective function, path flows, link flows, link costs, and TSTT from their equilibrium values. Our approach is based on the partial linearization of the SUE objective, and the resulting bounds are computable with minimal computational effort. We show that the derived bounds are tight and cannot be reduced by a fixed factor uniformly over the class of problems we consider.
- We show that, near equilibrium, the error bounds on path flows, link flows, link costs, and TSTT converge to zero at least as fast as the distance between the current path flow and the equilibrium path flow, and the objective bound converges at least as fast as the square of this distance. Consequently, the derived bounds converge to zero at least as fast as the path flow iterates converge to equilibrium, regardless of the algorithm used to solve the stochastic TAP.

- We present detailed computational experiments on four city-level networks to evaluate the tightness of the derived bounds. At a relative gap of 10^{-6}, the bounds guarantee that the path flows, link flows, link costs, and TSTT are all within 0.1% of their equilibrium values for a logit dispersion parameter of 0.5, and within 1% for a dispersion parameter of 1.5. We then present a network design case study on the Chicago Sketch network, in which the bounds allow 96% of the candidate designs to be rejected before their traffic assignment is solved to high accuracy, reducing computation time by 45%.

The error bounds derived in this study have several practical applications. Quantities such as path flows, link flows, and TSTT can be reported as ranges rather than single values, with each range guaranteed to contain the equilibrium value even when the traffic assignment algorithm is terminated early. This can increase confidence in model outputs. Furthermore, these bounds can directly support pruning strategies in bi-level network design problems where the lower level is a stochastic TAP. For instance, suppose 10,000 stochastic TAP instances must be evaluated in a network design problem minimizing TSTT, and one candidate design is guaranteed to have an equilibrium TSTT of at most 30 units. Any other candidate whose lower bound on equilibrium TSTT exceeds 30 units can then be pruned by terminating its TAP early, as it cannot improve on the identified design. Another application is choosing a solution accuracy that reflects uncertainty in the input data. Solving for equilibrium to an accuracy much finer than warranted by the demand data adds computation without necessarily making the outputs more reliable. These bounds can help terminate the TAP once the guaranteed error in the quantities of interest falls below the estimated uncertainty in those quantities due to input errors.

The remainder of the paper is organized as follows. Section 2 reviews the literature relevant to the problem. Section 3 presents the logit route choice model and the key mathematical properties of SUE required to develop the bounds. Section 4 presents the main contributions of the paper, namely upper bounds on the deviations of path flows, link flows, link costs, and TSTT from their equilibrium values. Section 5 analyzes the convergence of these bounds as the current solution approaches equilibrium. Section 6 presents the computational results on the tightness of the bounds and a network design case study. Finally, Section 7 summarizes the findings and presents directions for future work.

2 Literature review

Static TAPs are route choice models that predict user behavior under varying congestion levels, given fixed demands between origin-destination (OD) pairs in a network. Stochastic TAPs, a class of static TAPs, assume that users do not have perfect knowledge of travel costs and that their perceptions of these costs are subject to errors (Daganzo and Sheffi 1977). A route choice model then specifies how these errors are distributed across different routes available to a user. Prashker and Bekhor (2004) provided an extensive review of several such route choice models. Among these, the multinomial logit model assumes that the errors are independently and identically distributed (iid) Gumbel random variables. It is a widely used route choice model because the resulting route choice probabilities have a closed-form expression and the equilibrium can be characterized as the minimizer of a strictly convex objective (Fisk 1980). However, the iid assumption may be restrictive when paths overlap, since shared links can induce correlation in the unobserved utilities of these paths. The probit model relaxes this assumption by allowing multivariate normal errors (Daganzo and Sheffi 1977). However, its route choice probabilities do not have a closed-form expression and are therefore typically evaluated through simulation.

Several route choice models extend the logit choice model to capture correlation between overlapping paths while still preserving analytical tractability. These include C-logit (Cascetta et al. 1996, Zhou et al. 2012), implicit availability/perception logit (Cascetta et al. 1999), path-size logit (Ben-Akiva and Bierlaire 1999), paired combinatorial logit (Chu 1989), cross-nested logit (Vovsha 1997), recursive logit (Fosgerau et al. 2013), and the perturbed utility choice model (Fosgerau et al. 2022, Yao et al. 2024, Yao

and Zhang 2026, Nie 2026). In this paper, we focus only on the multinomial logit model. Extending our derived bounds to these broader formulations would be valuable but is beyond the scope of this paper.

Algorithms used to solve for logit-SUE span across link-based and path-based methods. Link-based methods include the convex combination algorithm of Chen and Alfa (1991), the optimal step length algorithm of Maher (1998), and the self-regulated scheme of Liu et al. (2009), all of which build on the traditional method of successive averages. Path-based methods include disaggregate simplicial decomposition (Damberg et al. 1996), disaggregate simplicial decomposition with a self-adaptive Armijo line search (Chen et al. 2013), fixed-point iterations with Barzilai-Borwein step sizes (Du et al. 2021), gradient projection (Bekhor and Toledo 2005), gradient projection with Barzilai-Borwein step sizes (Wang et al. 2025), and Newton's method (Zhou et al. 2015, Xu et al. 2019, Bagchi and Boyles 2026). The path-based methods are generally computationally faster but require a specified path set. Path sets can be chosen in several ways. These include methods that enumerate a finite set of paths (such as k-shortest paths and link-penalty methods) and those that operate over an implicit path set without generating every path (such as the methods of Dial (1971) and Bell (1995)). In our analysis, we focus on developing error bounds for path-based algorithms and assume that each OD pair has a fixed, finite set of acyclic paths.

All the algorithms discussed above are iterative and only approach equilibrium in the limit. Traffic assignment is a general non-linear programming problem, so its exact solution generally cannot be found in a finite amount of time (LeBlanc et al. 1975). These procedures must therefore be terminated before reaching exact equilibrium, raising questions about when to stop and how close the final solution is to equilibrium. One of the earliest studies to examine these questions was by Rose et al. (1988), who empirically investigated the factors determining the magnitude of convergence errors in the Frank-Wolfe algorithm for DUE. In particular, they noted that stopping criteria "do not directly indicate how close the predictions are to the true equilibrium flows." Later, Boyce et al. (2004) suggested that a relative gap of 10^{-4} was needed to stabilize link flows in DUE, leading to its widespread adoption as a stopping criterion. More recently, Patil et al. (2021) provided a more detailed analysis of the convergence of several aggregate metrics in deterministic traffic assignment. Their analysis showed that metrics at higher levels of aggregation stabilized earlier, at larger relative gap values. For instance, they found that TSTT and vehicle miles traveled stabilized at a relative gap of around 10^{-4}, link flows at around 10^{-5}, and path flows at around 10^{-6}.

In non-linear optimization, duality-based gaps are often used since they directly bound the error in the corresponding objective function. Hearn (1982) defined a duality gap function for a convex program and showed that the difference of the objective function evaluated at any solution and the duality gap gives a lower bound on the optimal objective. To obtain this lower bound, the objective is linearized at the current solution and minimized over the feasible set, and the gap is computed as the difference between the current objective and this minimum. When applied to DUE, this gap function equals the difference between the TSTT and the shortest path travel time, defined as the total travel time if all trips used minimum-time paths computed at the current flows.

When the objective function is the sum of a convex function and a differentiable function, a partial linearization can be used, in which only the differentiable part is linearized and the convex part is kept exact (Patriksson 1993). The SUE objective of Fisk (1980) is well suited to this approach as it is the sum of a differentiable cost component and a convex entropy component. It turns out that the minimum of this partially linearized objective is attained at the logit flows at the current path costs (Leurent 1997). Leurent (1997) used this idea to derive a duality gap for the SUE objective to obtain what the author calls a "theoretically sound convergence test" for the method of successive averages. In doing so, the author developed a lower bound on the optimal SUE objective. However, this lower bound was used only as a stopping criterion and not interpreted as an error bound on the objective. Therefore, its tightness was not examined, nor was it extended to error bounds on flows and aggregate metrics such as TSTT. Damberg et al. (1996) also noted, without proof, that for a given set of routes, the optimal

value of the partially linearized SUE problem is a lower bound on the optimal SUE objective, so that a criterion based on the relative objective error can be used for termination. However, their main goal was to provide an algorithm for logit-based SUE, and they did not extend this observation to error bounds on flows or aggregate metrics. In this article, we first restate this lower bound as an error bound on the SUE objective at the current solution, and show that it can be computed with little additional effort from the path costs and logit flows that many iterative procedures already compute at each iteration. We then extend the error bound on the SUE objective to derive error bounds on path flows, link flows, link costs, and TSTT.

The idea of bounding the distance between an approximate and an exact solution using a gap measure computed only from the current iterate is not new. Pang (1987) derived such a posteriori error bounds for variational inequalities with strongly monotone mappings, which for optimization problems correspond to strongly convex objectives. Logit-SUE can be written as a variational inequality through the optimality conditions of Fisk's program (Guo et al. 2010), and Fisk's objective is strongly convex over the feasible path flows. However, for this variational inequality, the gap function of Pang (1987) amounts to linearizing the entire objective, including the entropy term, whose gradient is unbounded as path flows approach zero. In this article, we instead use the partial linearization discussed above, which keeps the entropy term exact. This avoids the issue with zero path flows, and as we show, provides a tighter bound on the SUE objective than the full linearization. We then use the strong convexity of the SUE objective to carry the bound on the SUE objective over to path flows, measuring the path flow error within each OD pair in a demand-weighted norm.

To the best of our knowledge, no prior study in traffic assignment has theoretically derived error bounds in flows or in aggregate metrics such as TSTT that can be computed at any feasible solution. Bagchi (2025) explored local bounds based on Taylor's theorem applied to the fixed-point formulation of logit-SUE, but these bounds apply only within a neighborhood of the equilibrium that cannot be verified at a given iterate. Consequently, error bounds have not been used to decide when to terminate a stochastic TAP based on the accuracy required in the metric of interest, which would complement the empirical evidence that prior studies used to choose stopping criteria for TAPs. We address this gap by deriving error bounds that build on ideas from the non-linear optimization literature and strengthen them using properties specific to the TAP. We then present computational results on four city-level networks to test the tightness of these bounds, along with a network design case study to demonstrate their practical use.

3 Background

We now present some background on stochastic TAP along with the notation conventions we use throughout this article. We denote scalar-valued functions by lowercase Greek letters (e.g., γ), vector-valued functions by calligraphic letters (e.g., $\mathcal{L}$), and functions from scalars to vectors by script letters (e.g., $\mathscr{h}$). Sets are represented by blackboard-bold letters (e.g., $\mathbb{H}$). Vectors are written as bold lowercase letters (e.g., $\boldsymbol{h}$) and matrices as bold uppercase letters (e.g., $\boldsymbol{D}$). All other scalars are written in Latin letters (e.g., d_w), except the dispersion parameter θ. Table 1 summarizes the notation used in this article, and Table 2 lists several "shorthands" we use to enhance the readability of our formulations.

We denote the set of links in a network by $\mathbb{L}$, with each link indexed by l, and denote a path by i. The set of OD pairs is denoted by $\mathbb{W}$, with a generic OD pair w having fixed positive demand d_w. The maximum demand across all OD pairs is denoted as $d_{\max}$. For each OD pair w, $\mathbb{P}_w$ denotes a given set of acyclic paths connecting it. The union of these sets is denoted by $\mathbb{P}$, with $n = |\mathbb{P}|$ the total number of paths. We assume that the set of paths $\mathbb{P}$ is fixed and does not change during the assignment procedure. Although we assume our path set to contain only acyclic paths in this article, our results extend to any finite path set, including those containing cycles, with minimal modifications. We discuss these modifications in Section 7.

Table 1: Model notation.

Symbol	Dimension / range	Meaning
$\mathbb{L}$	–	Set of links.
$\mathbb{H}, \mathbb{A}$	–	Sets of feasible path flows and link flows.
$w \in \mathbb{W}$	–	OD pair w in set of OD pairs.
$\mathbb{P}, \mathbb{P}_w$	$\lvert\mathbb{P}\rvert = n$, $\lvert\mathbb{P}_w\rvert = n_w$	Set of all paths and paths serving OD pair w.
l, i, j	Indices	Link index (l), path indices (i, j).
$w(i)$	–	OD pair served by path i.
$\boldsymbol{h}, \boldsymbol{h}^w$	$\mathbb{R}^n, \mathbb{R}^{n_w}$	Vector of current path flows and its OD sub-vector.
$\hat{\boldsymbol{h}}, \hat{\boldsymbol{h}}^w$	$\mathbb{R}^n, \mathbb{R}^{n_w}$	Vector of equilibrium path flows and its OD sub-vector.
$\boldsymbol{a}, \hat{\boldsymbol{a}}$	$\mathbb{R}^{\lvert\mathbb{L}\rvert}$	Vector of current and equilibrium link flows.
$\boldsymbol{h}^{(k)}, \boldsymbol{a}^{(k)}$	$\mathbb{R}^n, \mathbb{R}^{\lvert\mathbb{L}\rvert}$	kth iterate of path flows and corresponding link flows.
$\boldsymbol{q}$	$\mathbb{R}^n$	Current marginal social path costs, $\boldsymbol{D}^T \nabla\zeta(\boldsymbol{a})$.
d_w	> 0	Demand of OD pair w.
θ	> 0	Logit dispersion parameter.
$\boldsymbol{D}, D_{li}$	$\{0,1\}^{\lvert\mathbb{L}\rvert \times n}$	Link-path incidence matrix and its entries.
τ_l	$\mathbb{R} \to \mathbb{R}$	Link performance function.
$\mathcal{T}$	$\mathbb{R}^{\lvert\mathbb{L}\rvert} \to \mathbb{R}^{\lvert\mathbb{L}\rvert}$	Link-cost mapping.
$\mathcal{C}$	$\mathbb{R}^n \to \mathbb{R}^n$	Path-cost mapping.
$\mathcal{L}$	$\mathbb{R}^n \to \mathbb{R}^n$	Logit target-flow mapping.
$L_{\mathcal{L}}$	≥ 0	Lipschitz constant of $\mathcal{L}$ on $\mathbb{H}$.
ϕ, β, η	$\mathbb{R}^n_{\geq 0} \to \mathbb{R}$	SUE objective, cost component, and entropy component.
ζ	$\mathbb{R}^{\lvert\mathbb{L}\rvert} \to \mathbb{R}$	Total system travel time, $\zeta(\boldsymbol{a}) = \boldsymbol{a}^T \mathcal{T}(\boldsymbol{a})$.
$\lVert\cdot\rVert, \lVert\cdot\rVert_1$	Norms	ℓ_2 norm and ℓ_1 norm.

For a path i, we write $w(i)$ for the OD pair to which path i belongs. The vector of link flows is denoted by $\boldsymbol{a}$, where a_l is the flow on link l. The vector of path flows in an OD pair w is denoted as $\boldsymbol{h}^w$, and we write $\boldsymbol{h}$ as the ordered concatenation of the sub-vectors $\boldsymbol{h}^w$. We say a path flow vector is *feasible* if it belongs to the set

$$\mathbb{H} := \left\{ \boldsymbol{h} \geq \boldsymbol{0} : \sum_{i \in \mathbb{P}_w} h_i = d_w \quad \forall w \in \mathbb{W} \right\}. \tag{1}$$

The link-path incidence matrix, denoted $\boldsymbol{D}$, is defined by

$$D_{li} = \begin{cases} 1, & \text{if link } l \text{ lies on path } i, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

The set of feasible link flows is the image of $\mathbb{H}$ produced by the action of the matrix $\boldsymbol{D}$:

$$\mathbb{A} := \{\boldsymbol{D}\boldsymbol{h} : \boldsymbol{h} \in \mathbb{H}\}. \tag{3}$$

It is well known that the sets $\mathbb{H}$ and $\mathbb{A}$ are both compact and convex.

The path costs for a given path flow vector are computed using network loading. Given a feasible path flow vector, we first obtain the link flows as $\boldsymbol{a} = \boldsymbol{D}\,\boldsymbol{h}$ and the corresponding link costs using the link performance function $\tau_l(a_l)$. We assume link costs are separable, twice continuously differentiable, and monotone non-decreasing on $[0, \infty)$. Let $\mathcal{T}(\boldsymbol{a})$ be the vector of link costs, where $\mathcal{T}_l(\boldsymbol{a}) = \tau_l(a_l)$. The path cost mapping, defined as $\mathcal{C}(\boldsymbol{h}) = \boldsymbol{D}^T\,\mathcal{T}(\boldsymbol{a}) = \boldsymbol{D}^T\,\mathcal{T}(\boldsymbol{D}\boldsymbol{h})$, then computes the path cost as the sum of the costs of the links it contains.

SUE accounts for travelers' imperfect perception of travel costs by assuming that route choices are based on perceived rather than actual travel costs. If the perception errors are iid Gumbel random variables, the resulting route choice behavior follows the logit model. In this case, the target path flows can be expressed in closed form as a continuously differentiable function of the current path flows.

Given a feasible path flow vector, the logit mapping gives the target flow on each path as

$$\mathcal{L}_i(\boldsymbol{h}) := \frac{d_{w(i)} \exp[-\theta \mathcal{C}_i(\boldsymbol{h})]}{\sum_{j \in \mathbb{P}_{w(i)}} \exp[-\theta \mathcal{C}_j(\boldsymbol{h})]}, \qquad i \in \mathbb{P}, \tag{4}$$

where $\theta > 0$ is the dispersion parameter. The mapping $\mathcal{L}$ is continuously differentiable under the assumed link performance functions. Furthermore,

$$\mathcal{L}(\boldsymbol{h}) > \mathbf{0} \quad \forall \boldsymbol{h} \in \mathbb{H}. \tag{5}$$

This is because link performance functions are finite on the compact set $\mathbb{A}$, so every path cost $\mathcal{C}_i(\boldsymbol{h})$ is finite. Equation (4) then gives $\mathcal{L}_i(\boldsymbol{h}) > 0$ for every $i \in \mathbb{P}$.

We denote the equilibrium path flow vector by $\hat{\boldsymbol{h}}$, which satisfies $\hat{\boldsymbol{h}} = \mathcal{L}(\hat{\boldsymbol{h}})$, and write $\hat{\boldsymbol{a}} = \boldsymbol{D}\hat{\boldsymbol{h}}$ for the equilibrium link flows. Fisk (1980) showed that the SUE under these assumptions can be characterized as the unique minimizer of a strictly convex program, with the objective:

$$\begin{aligned} \phi(\boldsymbol{h}) &:= \beta(\boldsymbol{h}) + \eta(\boldsymbol{h}), \\ \beta(\boldsymbol{h}) &:= \sum_{l \in \mathbb{L}} \int_0^{(\boldsymbol{D}\boldsymbol{h})_l} \tau_l(x)\, dx, \\ \eta(\boldsymbol{h}) &:= \frac{1}{\theta} \sum_{i \in \mathbb{P}} h_i \ln h_i, \end{aligned} \tag{6}$$

and the convention $0 \ln 0 := 0$.

Table 2: Shorthands.

Symbol	Formula
$\beta_h^{\text{lin}}(\tilde{\boldsymbol{h}})$	$\beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T(\tilde{\boldsymbol{h}} - \boldsymbol{h})$
$\psi_h(\tilde{\boldsymbol{h}})$	$\beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T(\tilde{\boldsymbol{h}} - \boldsymbol{h}) + \eta(\tilde{\boldsymbol{h}})$
$\gamma(\boldsymbol{h})$	$\psi_h(\boldsymbol{h}) - \psi_h(\mathcal{L}(\boldsymbol{h}))$
$d_{\max}$	$\max_{w \in \mathbb{W}} d_w$
$\boldsymbol{v}, \boldsymbol{v}^w$	$\boldsymbol{h} - \hat{\boldsymbol{h}}, \quad \boldsymbol{h}^w - \hat{\boldsymbol{h}}^w$
$\hbar(t)$	$\hat{\boldsymbol{h}} + t\boldsymbol{v}, \quad 0 \le t \le 1$
$\mathscr{g}(t)$	$\mathcal{L}(\boldsymbol{h}^{(k)}) + t\big(\boldsymbol{h}^{(k)} - \mathcal{L}(\boldsymbol{h}^{(k)})\big), \quad 0 \le t \le 1$
m_h	$\sqrt{\theta d_{\max} \gamma(\boldsymbol{h})}$
$\mathbb{W}_l$	$\left\{ w \in \mathbb{W} :\ 0 < \sum_{i \in \mathbb{P}_w} D_{li} < \lvert\mathbb{P}_w\rvert \right\}$
S_l	$\sum_{w \in \mathbb{W}_l} d_w$
$\tilde{m}_a^l$	$\sqrt{\theta S_l \gamma(\boldsymbol{h})/2}$
R_a^l	$\min\{\tau_l'(\tilde{a}_l) :\ \tilde{a}_l \ge 0,\ \lvert\tilde{a}_l - a_l\rvert \le \tilde{m}_a^l\}$
m_a^l	$\tilde{m}_a^l / \sqrt{1 + \theta S_l R_a^l / 4}$
$\mathbb{A}_a^l$	$\{x \in \mathbb{R}_{\ge 0} : \lvert x - a_l\rvert \le m_a^l\}$
L_a^l	$\max_{\tilde{a}_l \in \mathbb{A}_a^l} \tau_l'(\tilde{a}_l)$
K_a^l	$\max_{\tilde{a}_l \in \mathbb{A}_a^l} \lvert 2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l)\rvert$
Q_a	$\sqrt{\sum_{w \in \mathbb{W}} d_w \left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i\right)^2}$

The TSTT is represented by the scalar-valued function

$$\zeta(\boldsymbol{a}) := \boldsymbol{a}^T \mathcal{T}(\boldsymbol{a}). \tag{7}$$

Since link costs are assumed to be separable, differentiating ζ with respect to the link flows gives

$$[\nabla \zeta(\boldsymbol{a})]_l = \tau_l(a_l) + a_l \tau_l'(a_l), \qquad l \in \mathbb{L}. \tag{8}$$

Applying the chain rule, we define the vector of marginal social path costs at the current flows as

$$\boldsymbol{q} := \nabla_{\boldsymbol{h}}[\zeta(\boldsymbol{D}\boldsymbol{h})] = \boldsymbol{D}^T \nabla \zeta(\boldsymbol{a}). \tag{9}$$

Here, q_i measures the rate of change in TSTT when flow is added to path i, holding the other path flows fixed.

Our goal in this article is to find bounds on the distance to equilibrium from any given feasible flow. Throughout this article, we use m_h for a bound on path flow errors ($\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$), and m_a^l for a bound on link flow errors ($|a_l - \hat{a}_l|$). The link flow bound defines an interval around the current link flow:

$$\mathbb{A}_a^l := \{x \in \mathbb{R}_{\geq 0} : |x - a_l| \leq m_a^l\}, \qquad l \in \mathbb{L}. \tag{10}$$

We will characterize this interval at the current solution so that it is guaranteed to contain the equilibrium link flow $\hat{a}_l$. We will then use this interval to determine bounds on link costs and TSTT.

4 Bound formulations

For any feasible path flow solution, we start by bounding the difference between the SUE objective evaluated at that solution and the optimal SUE objective in Section 4.1. We then use this result to obtain path flow bounds in Section 4.2. In particular, we will show that the path flow error is upper bounded by a constant times the square root of the difference in SUE objective values. Using the path flow error bound, we then derive error bounds on link flows and link costs in Section 4.3, and on TSTT in Section 4.4. Throughout this section, "current path flow" will denote any feasible path flow solution to the stochastic TAP, such as an iterate of an algorithm solving for SUE.

4.1 Bounds on the objective

We recall that the SUE objective, ϕ, is the sum of two components: β, the cost component, and η, the entropy component. To derive a bound on the SUE objective, we will use the fact that β is convex and hence lies above its tangent. By replacing β with its tangent at the current solution, we will obtain a lower approximation of the SUE objective that agrees with it at the current flows. We will then minimize this approximation to obtain a lower bound on the equilibrium objective. This idea is not particularly new; Leurent (1997) used the same argument to derive a lower bound on the SUE objective. However, we write the resulting theorem as an error bound, which defines a gap measure $\gamma(\boldsymbol{h})$ at any feasible path flow $\boldsymbol{h}$, and then connect this gap to error bounds on path flows, link flows, link costs, and TSTT. Deriving these bounds using $\gamma(\boldsymbol{h})$ remains the main contribution of this article.

We write β_h^{lin} for the tangent approximation to β at $\boldsymbol{h}$. For a fixed feasible current path flow solution $\boldsymbol{h}$, we then define the partially linearized SUE objective

$$\psi_h(\tilde{\boldsymbol{h}}) := \beta_h^{\text{lin}}(\tilde{\boldsymbol{h}}) + \eta(\tilde{\boldsymbol{h}}). \tag{11}$$

The subscript in the definition of ψ indicates that we hold the current flows $\boldsymbol{h}$ fixed while varying $\tilde{\boldsymbol{h}}$. Differentiating β with respect to current path flows gives

$$\nabla \beta(\boldsymbol{h}) = \boldsymbol{D}^T \mathcal{T}(\boldsymbol{D}\boldsymbol{h}) = \mathcal{C}(\boldsymbol{h}). \tag{12}$$

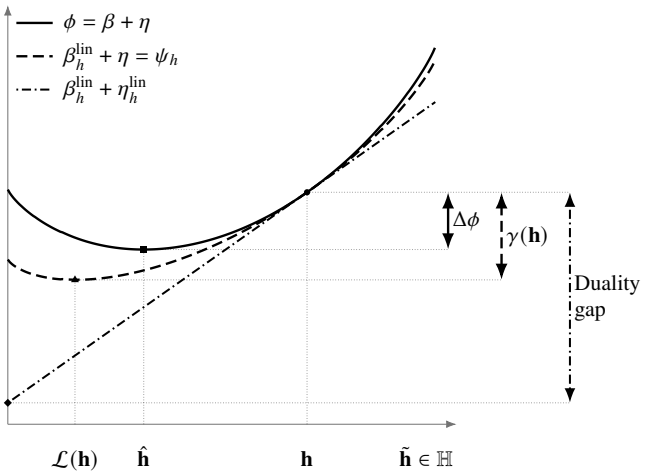


Figure 1: Intuition behind the SUE objective bound.

The solid curve represents the SUE objective, the dashed curve represents the partially linearized objective ψ_h, and the dash-dot line represents the full linear approximation to ϕ at $\boldsymbol{h}$. The vertical gaps show the true error in the SUE objective, the error bound used in this study, and the error bound obtained from the duality gap of the objective.

Using the gradient of β from Equation (12) to linearize β in Equation (11) then gives

$$\psi_h(\tilde{\boldsymbol{h}}) = \beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T(\tilde{\boldsymbol{h}} - \boldsymbol{h}) + \eta(\tilde{\boldsymbol{h}}). \tag{13}$$

It is well known that the unique minimizer of ψ_h over the feasible set $\mathbb{H}$ is $\mathcal{L}(\boldsymbol{h})$ (Leurent 1997, Bekhor and Toledo 2005). For completeness, we also prove this result in Lemma 5 in Appendix A.

Figure 1 illustrates our proof technique. The dashed curve represents $\psi_h = \beta_h^{\text{lin}} + \eta$ as a lower approximation of the SUE objective at $\boldsymbol{h}$. Since the minimizer of ψ_h over the feasible set is $\mathcal{L}(\boldsymbol{h})$, the difference between $\psi_h(\boldsymbol{h})$ and $\psi_h(\mathcal{L}(\boldsymbol{h}))$ upper bounds the difference in objective values at $\boldsymbol{h}$ and $\hat{\boldsymbol{h}}$. Figure 1 also illustrates why retaining η helps when linearizing ϕ. If we linearize the full objective, we replace η by its tangent as well, giving another lower bound on the equilibrium objective. However, this bound can never be better than the one obtained by retaining η. Since η is convex, it lies above its tangent, so ψ_h lies above the full linearization over the feasible set, and the minimum of ψ_h is therefore at least the minimum of the full linearization. We now state the result formally in Theorem 1.

Theorem 1 (Error bounds on the SUE objective)**.** *For any feasible path flow vector* $\boldsymbol{h}$*,*

$$\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) \leq \gamma(\boldsymbol{h}) := \psi_h(\boldsymbol{h}) - \psi_h(\mathcal{L}(\boldsymbol{h})). \tag{14}$$

Proof. By convexity of β,

$$\beta(\hat{\boldsymbol{h}}) \geq \beta_h^{\text{lin}}(\hat{\boldsymbol{h}}) = \beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T(\hat{\boldsymbol{h}} - \boldsymbol{h}). \tag{15}$$

Adding $\eta(\hat{\boldsymbol{h}})$ to both sides, we obtain

$$\begin{aligned} \beta(\hat{\boldsymbol{h}}) + \eta(\hat{\boldsymbol{h}}) &\geq \beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T(\hat{\boldsymbol{h}} - \boldsymbol{h}) + \eta(\hat{\boldsymbol{h}}) \\ &= \psi_h(\hat{\boldsymbol{h}}). \end{aligned} \tag{16}$$

Since $\phi = \beta + \eta$, this gives

$$\phi(\hat{\boldsymbol{h}}) \geq \psi_h(\hat{\boldsymbol{h}}). \tag{17}$$

We now use the fact that $\mathcal{L}(\boldsymbol{h})$ minimizes ψ_h over the feasible set (Lemma 5) to obtain

$$\psi_h(\hat{\boldsymbol{h}}) \geq \psi_h(\mathcal{L}(\boldsymbol{h})). \tag{18}$$

Combining inequalities (17) and (18), we obtain

$$\phi(\hat{\boldsymbol{h}}) \geq \psi_h(\mathcal{L}(\boldsymbol{h})). \tag{19}$$

Finally, subtracting the lower bound on $\phi(\hat{\boldsymbol{h}})$ from $\phi(\boldsymbol{h})$ gives

$$\begin{aligned} \phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) &\leq \phi(\boldsymbol{h}) - \psi_h(\mathcal{L}(\boldsymbol{h})) \\ &= \psi_h(\boldsymbol{h}) - \psi_h(\mathcal{L}(\boldsymbol{h})) \\ &= \gamma(\boldsymbol{h}). \end{aligned} \tag{20}$$

The second step follows because, at the current flows, the definition of ψ_h gives $\psi_h(\boldsymbol{h}) = \beta(\boldsymbol{h}) + \eta(\boldsymbol{h}) = \phi(\boldsymbol{h})$. This proves the result. □

4.2 Bounds on path flows

With Theorem 1 bounding the difference in objective values, our next goal is to use this result to bound the distance between the current and equilibrium path flows. To this end, we will first show that the squared path flow error is upper bounded by a constant times the difference in SUE objective values in Lemma 1. In this process, we will apply Taylor's theorem along the segment joining the current path flows to the equilibrium path flows, and bound the second-order terms in the expansion. Combining Lemma 1 with Theorem 1 will then give us the path flow error bounds in Theorem 2.

Lemma 1. *For every feasible path flow vector* $\boldsymbol{h}$*,*

$$\frac{1}{2\theta} \sum_{w \in \mathbb{W}} \frac{\|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1^2}{d_w} \leq \phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}). \tag{21}$$

Proof. We will show the result along the line segment from $\hat{\boldsymbol{h}}$ to any fixed feasible $\boldsymbol{h}$. Writing $\boldsymbol{v} := \boldsymbol{h} - \hat{\boldsymbol{h}}$, we define

$$\hbar(t) := \hat{\boldsymbol{h}} + t\boldsymbol{v}, \qquad 0 \leq t \leq 1. \tag{22}$$

The set of feasible path flows is convex, so every $\hbar(t)$ is feasible and $\sum_{i \in \mathbb{P}_w} \hbar_i(t) = d_w$ for every OD pair w. Furthermore, since $\hat{\boldsymbol{h}} = \mathcal{L}(\hat{\boldsymbol{h}}) > \boldsymbol{0}$ (by Equation (5)), $\hbar_i(t) > 0$ for every $i \in \mathbb{P}$ and every $t \in [0, 1)$.

We first observe that the derivative of $\phi(\hbar(t))$ with respect to t is zero at the equilibrium solution, that is, at $t = 0$. This is because $\hat{\boldsymbol{h}} > \boldsymbol{0}$ and $\boldsymbol{v}$ preserves every OD demand, so both $\hat{\boldsymbol{h}} + t\boldsymbol{v}$ and $\hat{\boldsymbol{h}} - t\boldsymbol{v}$ are feasible for small positive t. Optimality of ϕ at $\hat{\boldsymbol{h}}$ therefore gives

$$\frac{d}{dt}\phi(\hbar(t))\bigg|_{t=0} = \nabla\phi(\hat{\boldsymbol{h}})^T \boldsymbol{v} = 0. \tag{23}$$

We now bound the second derivative of $\phi(\hbar(t))$ with respect to t. Since β is convex, its restriction to the line segment $\hbar(t)$ is convex, so

$$\frac{d^2}{dt^2}\beta(\hbar(t)) \geq 0. \tag{24}$$

For the entropy term, differentiating it twice gives, for $t \in [0, 1)$,

$$\frac{d^2}{dt^2}\eta(\hbar(t)) = \frac{1}{\theta} \sum_{w \in \mathbb{W}} \sum_{i \in \mathbb{P}_w} \frac{v_i^2}{\hbar_i(t)}. \tag{25}$$

To bound the right-hand side of Equation (25), we apply Titu's lemma (Sedrakyan and Sedrakyan 2018) separately within each OD pair,

$$\sum_{i\in\mathbb{P}_w} \frac{v_i^2}{\hbar_i(t)} \geq \frac{\left(\sum_{i\in\mathbb{P}_w} |v_i|\right)^2}{\sum_{i\in\mathbb{P}_w} \hbar_i(t)} = \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}. \tag{26}$$

Equation (25) along with Inequality (26) together give $\frac{d^2}{dt^2}\eta(\hbar(t)) \geq \frac{1}{\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}$. Adding this with Inequality (24), we get for every $t \in [0,1)$,

$$\begin{aligned} \frac{d^2}{dt^2}\phi(\hbar(t)) &= \frac{d^2}{dt^2}\beta(\hbar(t)) + \frac{d^2}{dt^2}\eta(\hbar(t)) \\ &\geq \frac{1}{\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}. \end{aligned} \tag{27}$$

Now using the integral form of Taylor's theorem, for $0 < s < 1$,

$$\begin{aligned} &\phi(\hbar(s)) - \phi(\hbar(0)) \\ &= s\,\frac{d}{dt}\phi(\hbar(t))\Big|_{t=0} + \int_0^s (s-t)\frac{d^2}{dt^2}\phi(\hbar(t))\,dt \\ &\geq \frac{1}{\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}\int_0^s (s-t)\,dt \\ &= \frac{s^2}{2\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}. \end{aligned} \tag{28}$$

Letting s approach one and using continuity of ϕ on $\mathbb{H}$ gives

$$\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) \geq \frac{1}{2\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1^2}{d_w}, \tag{29}$$

which proves the result. □

We are now ready to state the error bound on path flows. We will combine the upper bound on the SUE objective difference in Theorem 1 with the lower bound on the SUE objective difference in Lemma 1 to give a bound on the ℓ_1 path flow error. We will then use demand conservation within each OD pair to obtain an ℓ_2 bound for path flows across all OD pairs.

Theorem 2 (Error bounds on path flows)**.** *For every feasible path flow vector $\boldsymbol{h}$, the following bounds hold:*

1. *The ℓ_1 path flow errors satisfy*

$$\frac{1}{2\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1^2}{d_w} \leq \phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) \leq \gamma(\boldsymbol{h}). \tag{30}$$

   Consequently, for every OD pair $w \in \mathbb{W}$,

$$\|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1 \leq \sqrt{2\theta d_w \gamma(\boldsymbol{h})}. \tag{31}$$

2. *The ℓ_2 path flow error satisfies*

$$\|\boldsymbol{h} - \hat{\boldsymbol{h}}\| \leq m_h := \sqrt{\theta d_{\max}\gamma(\boldsymbol{h})}. \tag{32}$$

Proof. Combining Theorem 1 with Lemma 1 gives Equation (30). Since every term in the sum is non-negative, $\frac{\|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1^2}{d_w} \leq \sum_{w' \in \mathbb{W}} \frac{\|\boldsymbol{h}^{w'} - \hat{\boldsymbol{h}}^{w'}\|_1^2}{d_{w'}} \leq 2\theta\gamma(\boldsymbol{h})$. Multiplying by d_w and taking square roots gives Equation (31). This establishes the first part of the theorem.

To prove the second part, we write $\boldsymbol{v} := \boldsymbol{h} - \hat{\boldsymbol{h}}$. Since both path flow vectors meet the same OD demands, $\sum_{i \in \mathbb{P}_w} v_i = 0$. The positive and negative components of $\boldsymbol{v}^w$ therefore have equal total magnitude, each equal to $\|\boldsymbol{v}^w\|_1/2$. Hence,

$$\|\boldsymbol{v}^w\|^2 \leq \left(\sum_{\substack{i \in \mathbb{P}_w \\ v_i > 0}} v_i \right)^2 + \left(\sum_{\substack{i \in \mathbb{P}_w \\ v_i < 0}} |v_i| \right)^2 = \frac{1}{2} \|\boldsymbol{v}^w\|_1^2. \tag{33}$$

Using the result of the first part, we then obtain

$$\begin{aligned} \|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^2 &= \sum_{w \in \mathbb{W}} \|\boldsymbol{v}^w\|^2 \\ &\leq \frac{d_{\max}}{2} \sum_{w \in \mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w} \\ &\leq \theta d_{\max} \gamma(\boldsymbol{h}). \end{aligned} \tag{34}$$

Taking square roots establishes the result. □

4.3 Bounds on link flows and link costs

To derive an error bound on link flows and link costs, we will use the fact that a link flow is the sum of the path flows using that link and bound the resulting sum of path flow errors. To tighten the result, we will use demand conservation to remove OD pairs that contribute no error using the following fact. If all paths of an OD pair w use a link, the link flow contribution from that OD pair is equal to its demand d_w at both the current and equilibrium flows. On the other hand, if none of its paths use the link, the contribution is zero at both. Therefore, to bound the link flow errors on link l, we only need to consider the OD pairs that have at least one path using link l and at least one path not using it. We denote this set by

$$\mathbb{W}_l = \left\{ w \in \mathbb{W} : \; 0 < \sum_{i \in \mathbb{P}_w} D_{li} < |\mathbb{P}_w| \right\}, \tag{35}$$

for each $l \in \mathbb{L}$. We denote the sum of demand over this set of OD pairs for a link l by $S_l = \sum_{w \in \mathbb{W}_l} d_w$.

We will then show the link flow bounds can be strengthened further by including information on the slope of the link performance functions. We start by stating the error bounds on link flows by just summing the path flow errors in Lemma 2, and then strengthen this bound in Theorem 3. Finally, in Corollary 1, we use the link flow bounds to bound the errors in link costs.

Lemma 2. *For every $\boldsymbol{h} \in \mathbb{H}$ and each link $l \in \mathbb{L}$,*

$$|a_l - \hat{a}_l| \leq \tilde{m}_a^l := \sqrt{\frac{\theta S_l \gamma(\boldsymbol{h})}{2}}. \tag{36}$$

Proof. For a link l and an OD pair w, we consider two sets that together contain all paths of the OD pair: paths in OD pair w that use l and paths that do not use l. Since demand conservation implies

that the sum of path flow errors, $\sum_{i\in\mathbb{P}_w}(h_i-\hat{h}_i)=0$, the net path flow errors in these two sets must be negatives of each other. Hence,

$$\begin{aligned}\Big|\sum_{i\in\mathbb{P}_w} D_{li}(h_i-\hat{h}_i)\Big| &= \Big|\sum_{i\in\mathbb{P}_w}(1-D_{li})(h_i-\hat{h}_i)\Big| \\ &\le \frac{1}{2}\sum_{i\in\mathbb{P}_w}\big[D_{li}+(1-D_{li})\big]|h_i-\hat{h}_i| \\ &= \frac{1}{2}\|\boldsymbol{h}^w-\hat{\boldsymbol{h}}^w\|_1.\end{aligned} \tag{37}$$

Now, to obtain the link flow error, we need to add OD-specific errors over all OD pairs. We observe that for OD pairs $w\notin\mathbb{W}_l$, the contribution $\sum_{i\in\mathbb{P}_w} D_{li}(h_i-\hat{h}_i)$ is zero. This is because for every $w\notin\mathbb{W}_l$, either every path of w uses l, in which case the contribution equals $\sum_{i\in\mathbb{P}_w}(h_i-\hat{h}_i)=0$, or no path of w uses l, in which case $D_{li}=0$ for all i. Since $a_l-\hat{a}_l=\sum_{w\in\mathbb{W}}\sum_{i\in\mathbb{P}_w} D_{li}(h_i-\hat{h}_i)$, summing over the OD pairs in $\mathbb{W}_l$ and applying the Cauchy-Schwarz inequality gives

$$\begin{aligned}|a_l-\hat{a}_l| &\le \frac{1}{2}\sum_{w\in\mathbb{W}_l}\|\boldsymbol{h}^w-\hat{\boldsymbol{h}}^w\|_1 \\ &\le \frac{1}{2}\sqrt{\Big(\sum_{w\in\mathbb{W}_l} d_w\Big)\Big(\sum_{w\in\mathbb{W}_l}\frac{\|\boldsymbol{h}^w-\hat{\boldsymbol{h}}^w\|_1^2}{d_w}\Big)} \\ &\le \sqrt{\frac{\theta\gamma(\boldsymbol{h})}{2}S_l},\end{aligned} \tag{38}$$

where the last step uses Equation (30) from Theorem 2. □

The link flow bound can be tightened further. In Lemma 1, we used only the fact that β is convex, so that its second derivative is non-negative, and discarded this second derivative. We can instead keep the contribution to β from a single link l and bound it from below by the smallest value of τ_l' on the interval $\{x\ge 0:\ |x-a_l|\le\tilde{m}_a^l\}$, which contains the equilibrium link flow by Lemma 2.

As a result, the error in the SUE objective is lower bounded not only by a term involving the path flow errors but also by a term involving the link flow error on link l. From Theorem 1, the error in the SUE objective is upper bounded by $\gamma(\boldsymbol{h})$. So, using the same sandwich idea as in Theorem 2, $\gamma(\boldsymbol{h})$ upper bounds not just the path flow errors but the sum of a path flow error term and a link flow error term. Using the argument of Lemma 2, the path flow error term can in turn be bounded from below by the link flow error, so that both terms are expressed in the link flow error alone. Rearranging then gives a tighter link flow bound. We state this bound next and defer its proof to Appendix C for brevity.

Theorem 3 (Error bounds on link flows)**.** *For every feasible path flow vector $\boldsymbol{h}$ and each link $l\in\mathbb{L}$, define*

$$R_a^l := \min\Big\{\tau_l'(\tilde{a}_l):\ \tilde{a}_l\ge 0,\ |\tilde{a}_l-a_l|\le\tilde{m}_a^l\Big\}. \tag{39}$$

Then,

$$|a_l-\hat{a}_l|\le m_a^l := \frac{\tilde{m}_a^l}{\sqrt{1+\theta S_l R_a^l/4}}, \tag{40}$$

where $\tilde{m}_a^l$ is given by Lemma 2.

Since $R_a^l\ge 0$, the new bound is never larger than $\tilde{m}_a^l$. Computing R_a^l requires the minimum of τ_l' over the interval $\{x\ge 0:\ |x-a_l|\le\tilde{m}_a^l\}$. When τ_l is convex, as for the Bureau of Public Roads (BPR) link

performance functions, τ_l' is non-decreasing, so the minimum is attained at the left end of the interval and $R_a^l = \tau_l'\big(\max\{0,\, a_l - \tilde{m}_a^l\}\big)$, which requires a single evaluation of τ_l'.

Theorem 3 implies that, at any current path flow solution, we can obtain an interval $\mathbb{A}_a^l$, as defined in Section 3, that contains the equilibrium link flow $\hat{a}_l$. Because τ_l is assumed to be continuously differentiable, its derivative is bounded on this compact interval, so τ_l is Lipschitz continuous there with a constant equal to the largest value of τ_l' on $\mathbb{A}_a^l$. We call this Lipschitz constant L_a^l. As with R_a^l, if we assume τ_l is convex, as for BPR functions, $L_a^l = \tau_l'\big(a_l + m_a^l\big)$, which requires a single evaluation of τ_l'. Bounds on link costs then follow directly from Lipschitz continuity and the error bounds on link flows. We present this result in Corollary 1.

Corollary 1 (Error bounds on link costs)**.** *For $\boldsymbol{h} \in \mathbb{H}$, let m_a^l be given by Theorem 3, and define*

$$L_a^l := \max_{\tilde{a}_l \in \mathbb{A}_a^l} \tau_l'(\tilde{a}_l), \qquad l \in \mathbb{L}. \tag{41}$$

Then, for each link $l \in \mathbb{L}$,

$$|\tau_l(\hat{a}_l) - \tau_l(a_l)| \le L_a^l m_a^l. \tag{42}$$

Proof. Since τ_l' is continuous on the compact interval $\mathbb{A}_a^l$, it is bounded there by L_a^l. Furthermore, Theorem 3 gives $|\hat{a}_l - a_l| \le m_a^l$. Therefore, by the mean value theorem,

$$|\tau_l(\hat{a}_l) - \tau_l(a_l)| \le \max_{\tilde{a}_l \in \mathbb{A}_a^l} \tau_l'(\tilde{a}_l) |\hat{a}_l - a_l| \le L_a^l m_a^l, \tag{43}$$

which proves the result. □

4.4 Bound on TSTT

The path flow and link flow bounds can be extended to bound the error between the current and equilibrium TSTT. To this end, we will apply a first-order Taylor expansion to ζ at the current link flows $\boldsymbol{a}$ to estimate its value at equilibrium and then bound the resulting second-order error. We will then show that assuming the link performance functions are convex gives a tighter one-sided bound on equilibrium TSTT.

As with earlier proofs, we write $\boldsymbol{v} := \boldsymbol{h} - \hat{\boldsymbol{h}}$, so that $\hat{\boldsymbol{a}} - \boldsymbol{a} = -\boldsymbol{D}\boldsymbol{v}$. Applying Taylor's theorem to ζ around $\boldsymbol{a}$ gives

$$\begin{aligned} \zeta(\hat{\boldsymbol{a}}) - \zeta(\boldsymbol{a}) &= \nabla\zeta(\boldsymbol{a})^T(\hat{\boldsymbol{a}} - \boldsymbol{a}) \\ &\quad + \frac{1}{2}(\hat{\boldsymbol{a}} - \boldsymbol{a})^T \nabla^2\zeta(\tilde{\boldsymbol{a}})(\hat{\boldsymbol{a}} - \boldsymbol{a}), \end{aligned} \tag{44}$$

where $\tilde{\boldsymbol{a}}$ lies on the line segment between $\boldsymbol{a}$ and $\hat{\boldsymbol{a}}$.

Both terms in Equation (44) could be bounded using the link flow bounds in Theorem 3. However, bounding the link contributions independently ignores the fact that within an OD pair, flow removed from some paths must reappear on other paths. To retain this cancellation, we substitute $\hat{\boldsymbol{a}} - \boldsymbol{a} = -\boldsymbol{D}\boldsymbol{v}$. Using $\boldsymbol{q} = \boldsymbol{D}^T \nabla\zeta(\boldsymbol{a})$, the first-order term then becomes

$$\nabla\zeta(\boldsymbol{a})^T(\hat{\boldsymbol{a}} - \boldsymbol{a}) = -\nabla\zeta(\boldsymbol{a})^T \boldsymbol{D}\boldsymbol{v} = -\boldsymbol{q}^T\boldsymbol{v}. \tag{45}$$

Since ζ is separable over links, $\nabla^2\zeta(\tilde{\boldsymbol{a}})$ is diagonal with entries $2\tau_l'(\tilde{a}_l) + \tilde{a}_l\tau_l''(\tilde{a}_l)$, so the second-order term reduces to

$$\begin{aligned} &\frac{1}{2}(\hat{\boldsymbol{a}} - \boldsymbol{a})^T \nabla^2\zeta(\tilde{\boldsymbol{a}})(\hat{\boldsymbol{a}} - \boldsymbol{a}) \\ &\quad = \frac{1}{2}\sum_{l \in \mathbb{L}} \left[2\tau_l'(\tilde{a}_l) + \tilde{a}_l\tau_l''(\tilde{a}_l)\right](\hat{a}_l - a_l)^2, \end{aligned} \tag{46}$$

where each $\tilde{a}_l$ lies between a_l and $\hat{a}_l$. The remainder is quadratic in the link flow errors. We therefore use the link flow bounds for the remainder and exploit demand conservation to bound the first-order term only. We present the bounds on the first-order and second-order terms in Lemmas 3 and 4, respectively. The proofs of both are given in Appendix B.

Lemma 3. *For any feasible path flow vector $\boldsymbol{h}$, define*

$$Q_a := \sqrt{\sum_{w \in \mathbb{W}} d_w \left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i \right)^2}. \tag{47}$$

Then the first-order term in Equation (44) given by Equation (45) satisfies

$$|\boldsymbol{q}^T \boldsymbol{v}| \leq Q_a \sqrt{\frac{\theta \gamma(\boldsymbol{h})}{2}}. \tag{48}$$

Lemma 4. *For every feasible path flow vector $\boldsymbol{h}$, define*

$$K_a^l := \max_{\tilde{a}_l \in \mathbb{A}_a^l} \left| 2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right|, \qquad l \in \mathbb{L}. \tag{49}$$

Then the second-order term in Equation (44) given by Equation (46) satisfies

$$\begin{aligned} &\left| \frac{1}{2} \sum_{l \in \mathbb{L}} \left[2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right] (\hat{a}_l - a_l)^2 \right| \\ &\qquad \leq \frac{1}{2} \sum_{l \in \mathbb{L}} K_a^l (m_a^l)^2, \end{aligned} \tag{50}$$

where m_a^l is given by Theorem 3.

We are now ready to state the TSTT bound. Using Lemma 3 to bound the first-order term and Lemma 4 to bound the second-order term in Equation (44) gives the following result.

Theorem 4 (Error bounds on TSTT)**.** *For every feasible path flow vector $\boldsymbol{h}$,*

$$|\zeta(\hat{\boldsymbol{a}}) - \zeta(\boldsymbol{a})| \leq Q_a \sqrt{\frac{\theta \gamma(\boldsymbol{h})}{2}} + \frac{1}{2} \sum_{l \in \mathbb{L}} K_a^l (m_a^l)^2, \tag{51}$$

where Q_a, K_a^l, and m_a^l are given by Lemmas 3 and 4 and Theorem 3, respectively.

Proof. Substituting Equations (45) and (46) into Equation (44) gives

$$\begin{aligned} \zeta(\hat{\boldsymbol{a}}) - \zeta(\boldsymbol{a}) &= -\boldsymbol{q}^T \boldsymbol{v} \\ &\quad + \frac{1}{2} \sum_{l \in \mathbb{L}} \left[2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right] (\hat{a}_l - a_l)^2. \end{aligned} \tag{52}$$

Applying the triangle inequality to Equation (52), and then bounding the first term by Equation (48) in Lemma 3 and the second term by Equation (50) in Lemma 4 proves the result. □

We now assume that the link performance functions are convex. Together with monotonicity, the second-order term in the Taylor expansion of ζ becomes non-negative. We can therefore discard the second-order term to obtain a tighter lower bound on equilibrium TSTT. This gives us the following corollary.

Corollary 2 (Error bounds on TSTT for convex link performance functions)**.** *If each link performance function τ_l is convex on $[0, \infty)$, then for every feasible path flow vector $\boldsymbol{h}$,*

$$\zeta(\hat{\boldsymbol{a}}) \geq \zeta(\boldsymbol{a}) - Q_a \sqrt{\frac{\theta \gamma(\boldsymbol{h})}{2}}. \tag{53}$$

Proof. Monotonicity and convexity give $2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \geq 0$ for $\tilde{a}_l \geq 0$, and the second-order term in Equation (46) is therefore non-negative. Applying this bound in Equation (52), and then bounding only the first-order term $-\boldsymbol{q}^T \boldsymbol{v}$ gives

$$\zeta(\hat{\boldsymbol{a}}) \geq \zeta(\boldsymbol{a}) - \boldsymbol{q}^T \boldsymbol{v} \geq \zeta(\boldsymbol{a}) - Q_a \sqrt{\frac{\theta \gamma(\boldsymbol{h})}{2}}. \quad \square \tag{54}$$

We end this section by noting that the objective bound in Theorem 1 is tight when every link performance function is constant. In this case, β is linear in the path flows, so its tangent approximation at $\boldsymbol{h}$ is exact and hence $\psi_h = \phi$. Moreover, the path costs do not depend on the flows, so $\mathcal{L}(\boldsymbol{h}) = \mathcal{L}(\hat{\boldsymbol{h}}) = \hat{\boldsymbol{h}}$ for every feasible $\boldsymbol{h}$. Hence, $\gamma(\boldsymbol{h}) = \psi_h(\boldsymbol{h}) - \psi_h(\mathcal{L}(\boldsymbol{h})) = \psi_h(\boldsymbol{h}) - \psi_h(\hat{\boldsymbol{h}}) = \phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}})$, and the objective bound holds with equality. We further show in Appendix F that, for a family of two-link networks, the ratio of the error bounds derived in this study to the corresponding true error approaches one. Hence, none of the bounds derived in this study can be multiplied by a fixed factor smaller than one without further assumptions on the problem instance.

5 Convergence of the bounds

We now analyze the convergence of the bounds as the current solution approaches the SUE. To this end, we fix a network instance and consider a sequence of feasible solutions $\boldsymbol{h}^{(k)}$ approaching the SUE solution $\hat{\boldsymbol{h}}$, with link flows $\boldsymbol{a}^{(k)} = \boldsymbol{D}\boldsymbol{h}^{(k)}$. Our goal in this section is to show how the error bounds shrink with the distance between the current flows and their target flows, $\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|$. By relating this distance to the distance from the SUE solution, we can establish how fast the bounds approach zero as the path flow iterates approach equilibrium. Since $\mathcal{L}$ is continuously differentiable on the compact convex set $\mathbb{H}$, it is Lipschitz continuous there with some constant $L_{\mathcal{L}}$. Using $\mathcal{L}(\hat{\boldsymbol{h}}) = \hat{\boldsymbol{h}}$, we therefore have $\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\| \leq \|\mathcal{L}(\boldsymbol{h}^{(k)}) - \mathcal{L}(\hat{\boldsymbol{h}})\| + \|\hat{\boldsymbol{h}} - \boldsymbol{h}^{(k)}\| \leq (1 + L_{\mathcal{L}})\|\boldsymbol{h}^{(k)} - \hat{\boldsymbol{h}}\|$. Hence, the distance between the current flows and their target flows approaches zero at least as fast as the path flow iterates approach equilibrium.

Now, to establish the convergence of the error bounds, we will first show in Theorem 5 that $\gamma(\boldsymbol{h}^{(k)})$ approaches zero at least as fast as $\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|^2$. This will ensure that the path flow bounds derived in Theorem 2 approach zero at least as fast as $\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|$. We will then use this result in Corollary 3 to establish the same rate for the link flow, link cost, and TSTT bounds. Since $\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|$ approaches zero at least as fast as $\|\hat{\boldsymbol{h}} - \boldsymbol{h}^{(k)}\|$, these results show that the bounds derived in this study approach zero at least as fast as the iterates approach equilibrium, regardless of the solver used. The proofs of both Theorem 5 and Corollary 3 are deferred to Appendix D for brevity.

Theorem 5. *As $\boldsymbol{h}^{(k)} \to \hat{\boldsymbol{h}}$,*

$$\begin{aligned} \gamma(\boldsymbol{h}^{(k)}) &= O\Big(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|^2\Big), \\ m_{h^{(k)}} &= O\Big(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|\Big). \end{aligned} \tag{55}$$

Corollary 3. *As $\boldsymbol{h}^{(k)} \to \hat{\boldsymbol{h}}$, the link flow bounds m_a^l, the link cost bounds $L_a^l m_a^l$, and the TSTT bound in Theorem 4 are all $O(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|)$ and converge to zero.*

Table 3: Networks and path sets used in the study.

Network	OD pairs	Links	Nodes	Demand	Paths	$\overline{\text{CV}}$	$\overline{J}$
Sioux Falls	528	76	24	360600	10560	0.210	0.164
Anaheim	1406	914	416	104694	28120	0.064	0.455
Winnipeg	4344	2836	1052	64775	86880	0.027	0.659
Chicago Sketch	93135	2950	933	1137493	1862700	0.048	0.434

6 Computational results

The goal of this section is to present computational results to complement our theoretical developments. First, in Section 6.1, we test the validity and tightness of the bounds derived in Section 4, namely the bounds on the SUE objective, path flows, link flows, link costs, and TSTT. Then, in Section 6.2, we present a network design case study to demonstrate the practical applicability of the derived bounds. Additionally, in Appendix E, we examine how these bounds converge as the iterates approach equilibrium and report the computational cost of evaluating the bounds.

We tested the derived bounds on four real-world transportation networks from Transportation Networks for Research Core Team (2026): Sioux Falls, Anaheim, Winnipeg, and Chicago Sketch. We selected these networks because they have been widely used in SUE studies, including recent work by Zhou et al. (2015), Xu et al. (2019), Du et al. (2021), and Wang et al. (2025). For each OD pair, we generated the 20 shortest acyclic paths using Yen's k-shortest paths algorithm (Yen 1971). This choice of path sets is comparable to those used in recent SUE studies, such as Du et al. (2021) and Wang et al. (2025).

To assess the diversity of these path sets, we report two metrics. The first, $\overline{\text{CV}}$, is the coefficient of variation of free-flow path costs within each OD pair, averaged across all OD pairs. It measures how competitive the paths in our path sets are under free-flow conditions, with a lower value indicating more competitive paths. The second, $\overline{J}$, is the mean pairwise edge Jaccard index, which measures the extent to which paths overlap. The edge Jaccard index for two paths is the number of links they share divided by the number of links in their union. We computed $\overline{J}$ by averaging this index over all pairs of paths within each OD pair and then across all OD pairs. Lower values of $\overline{J}$ thus indicate that paths have lower overlap. Table 3 reports both these metrics for each network, together with the numbers of nodes, links, and OD pairs, the total demand, and the total number of paths in the generated path set.

In all our experiments, the stochastic TAP was solved using the algorithm of Du et al. (2021), which uses fixed-point iterations using the first Barzilai-Borwein (BB1) step size. We will refer to this algorithm as `FP-BB1` from now on. For our experiments, we used two values of the dispersion parameter, $\theta = 0.5$ and $\theta = 1.5$. These values lie within the ranges tested in recent studies on logit-SUE algorithms such as Du et al. (2021), who used $\theta \in [0.1, 1.6]$, and Wang et al. (2025), who used $\theta \in [0.2, 1.8]$. The initial solution was taken as the path flow vector obtained from logit network loading under free-flow conditions. As the convergence criterion, we used the RGAP for SUE (Du et al. 2021, Zhang et al. 2024) defined by

$$\begin{aligned} \text{RGAP}(\boldsymbol{h}) &= \frac{\sum_{i\in\mathbb{P}} h_i \big(g_i - \min_{j\in\mathbb{P}_{w(i)}} g_j\big)}{\sum_{i\in\mathbb{P}} h_i |g_i|}, \\ g_i &= \mathcal{C}_i(\boldsymbol{h}) + \frac{1}{\theta}\big(1 + \ln h_i\big). \end{aligned} \tag{56}$$

All our computational experiments were implemented in Python 3.12.5 and run on a Dell Precision 3680 workstation with an Intel Core i9-14900K processor and 128 GB of RAM. The source code of our experiments is available at `https://github.com/spartalab/sue-error-bounds`.

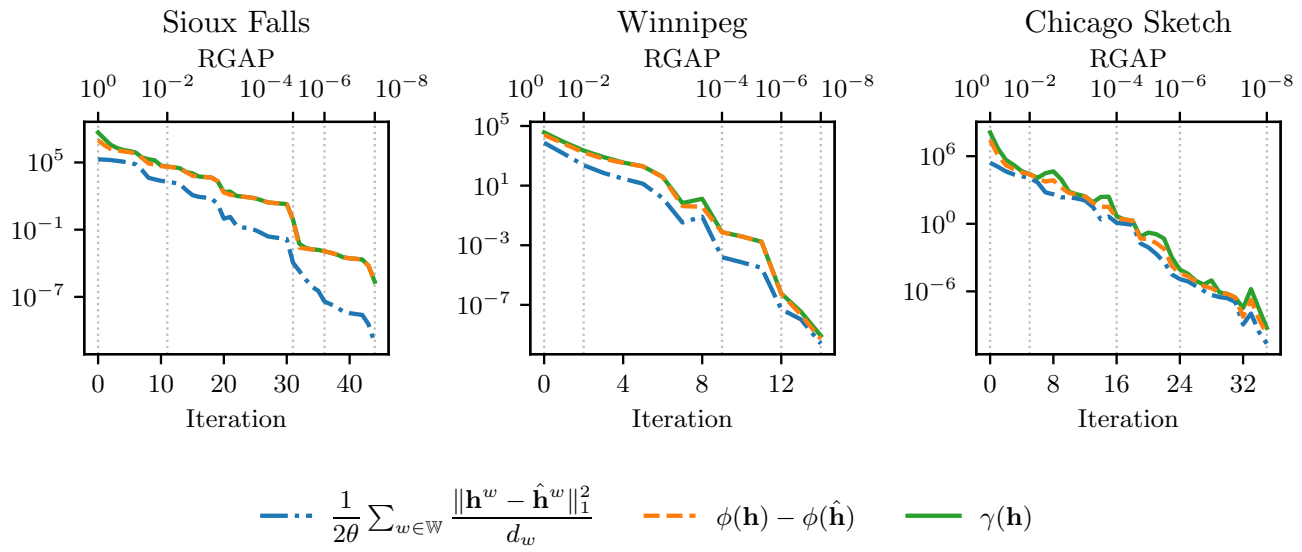


Figure 2: The three terms in Equation (57) versus iteration (log scale, $\theta = 0.5$).

6.1 Tightness of the bounds

To test the validity and tightness of the error bounds on the SUE objective, path flows, link flows, link costs, and TSTT, we compared them with the corresponding "true" errors relative to a reference equilibrium $\hat{\boldsymbol{h}}$. We obtained this reference equilibrium by running `FP-BB1` until the RGAP first fell below 10^{-12}. We then ran `FP-BB1` again from the same initial solution until the RGAP fell below 10^{-8}, recording the bounds and the corresponding errors relative to $\hat{\boldsymbol{h}}$ at each iteration.

We start by analyzing the objective bound in Theorem 1 and its tightness. Specifically, combining Lemma 1 with Theorem 1 shows that the difference in objective, $\Delta\phi$, is *sandwiched* between a demand-weighted sum of the squared path flow errors and the partially linearized gap:

$$\begin{aligned}\frac{1}{2\theta}\sum_{w\in\mathbb{W}}\frac{\|\boldsymbol{h}^w-\hat{\boldsymbol{h}}^w\|_1^2}{d_w} &\le \Delta\phi := \phi(\boldsymbol{h})-\phi(\hat{\boldsymbol{h}}) \\ &\le \gamma(\boldsymbol{h}).\end{aligned} \tag{57}$$

We plot this inequality chain for three networks, Sioux Falls, Winnipeg, and Chicago Sketch, in Figure 2. The figure shows the three terms in Equation (57) on a log scale as the iterations progress, along with the RGAP on the top axis. As we can observe, the upper bound $\gamma(\boldsymbol{h})$ on the objective error is reasonably tight in all three networks. However, the lower bound on the objective error is looser in the Sioux Falls network than in the Winnipeg and Chicago Sketch networks. This looseness stems from Inequality (27) in the proof of Lemma 1, where we discard the second derivative of β and bound the second derivative of η using Titu's lemma, both of which introduce slack into the inequality chain for realistic networks. Nevertheless, we observe that both bounds remain valid in all three networks at every iteration. To compare the strength of the bounds across different networks, we look at the errors as a percentage of the equilibrium value using the formulae shown in Table 4.

Table 5 reports the tightness of the bounds on the SUE objective gap from Theorem 1 and the path flow error from Theorem 2. It compares the actual objective difference $\Delta\phi$ and path flow error $\|\boldsymbol{h}-\hat{\boldsymbol{h}}\|$ with their respective upper bounds, $\gamma(\boldsymbol{h})$ and m_h, in the relative terms shown in Table 4. We present these relative errors at three RGAP values (10^{-4}, 10^{-6}, and 10^{-8}) for $\theta = 0.5$ and $\theta = 1.5$. The results indicate that the objective bound remains close to the actual difference in objective in all cases, while

Table 4: Definitions of true errors and bounds, expressed as percentages of equilibrium values.

	Objective	Path flow	Link flow	Link cost	TSTT
True %	$100\,\dfrac{\phi(\boldsymbol{h})-\phi(\hat{\boldsymbol{h}})}{\lvert\phi(\hat{\boldsymbol{h}})\rvert}$	$100\,\dfrac{\lVert\boldsymbol{h}-\hat{\boldsymbol{h}}\rVert}{\lVert\hat{\boldsymbol{h}}\rVert}$	$100\,\dfrac{\lVert\boldsymbol{a}-\hat{\boldsymbol{a}}\rVert}{\lVert\hat{\boldsymbol{a}}\rVert}$	$100\,\dfrac{\lVert\mathcal{T}(\boldsymbol{a})-\mathcal{T}(\hat{\boldsymbol{a}})\rVert}{\lVert\mathcal{T}(\hat{\boldsymbol{a}})\rVert}$	$100\,\dfrac{\lvert\zeta(\boldsymbol{a})-\zeta(\hat{\boldsymbol{a}})\rvert}{\zeta(\hat{\boldsymbol{a}})}$
Bound %	$100\,\dfrac{\gamma(\boldsymbol{h})}{\lvert\phi(\hat{\boldsymbol{h}})\rvert}$	$100\,\dfrac{m_h}{\lVert\hat{\boldsymbol{h}}\rVert}$	$100\,\dfrac{\lVert(m_a^l)_{l\in\mathbb{L}}\rVert}{\lVert\hat{\boldsymbol{a}}\rVert}$	$100\,\dfrac{\lVert(L_a^l m_a^l)_{l\in\mathbb{L}}\rVert}{\lVert\mathcal{T}(\hat{\boldsymbol{a}})\rVert}$	$100\,\dfrac{Q_a\sqrt{\theta\gamma(\boldsymbol{h})/2}+\frac{1}{2}\sum_{l\in\mathbb{L}}K_a^l(m_a^l)^2}{\zeta(\hat{\boldsymbol{a}})}$

Table 5: Objective and path flow errors and bounds, in percent of the equilibrium value.

		$\theta=0.5$				$\theta=1.5$			
		Objective		**Path flow**		**Objective**		**Path flow**	
Network	**RGAP**	**True %**	**Bound %**	**True %**	**Bound %**	**True %**	**Bound %**	**True %**	**Bound %**
Sioux Falls	10^{-4}	9×10^{-6}	9×10^{-6}	8×10^{-4}	2×10^{-1}	2×10^{-3}	2×10^{-3}	1×10^{-2}	4×10^{0}
	10^{-6}	1×10^{-8}	1×10^{-8}	1×10^{-5}	8×10^{-3}	1×10^{-6}	1×10^{-6}	5×10^{-5}	1×10^{-1}
	10^{-8}	3×10^{-11}	3×10^{-11}	2×10^{-7}	3×10^{-4}	3×10^{-8}	3×10^{-8}	5×10^{-7}	1×10^{-2}
Anaheim	10^{-4}	1×10^{-6}	3×10^{-6}	4×10^{-2}	4×10^{-1}	2×10^{-6}	3×10^{-6}	4×10^{-2}	4×10^{-1}
	10^{-6}	1×10^{-10}	1×10^{-10}	4×10^{-4}	3×10^{-3}	3×10^{-10}	3×10^{-10}	5×10^{-4}	4×10^{-3}
	10^{-8}	7×10^{-15}	1×10^{-14}	2×10^{-6}	3×10^{-5}	7×10^{-14}	8×10^{-14}	7×10^{-6}	6×10^{-5}
Winnipeg	10^{-4}	8×10^{-7}	8×10^{-7}	5×10^{-3}	2×10^{-1}	2×10^{-5}	2×10^{-5}	6×10^{-2}	2×10^{0}
	10^{-6}	6×10^{-11}	6×10^{-11}	1×10^{-4}	2×10^{-3}	7×10^{-7}	7×10^{-7}	1×10^{-3}	3×10^{-1}
	10^{-8}	5×10^{-14}	9×10^{-14}	5×10^{-6}	8×10^{-5}	1×10^{-9}	1×10^{-9}	1×10^{-5}	1×10^{-2}
Chicago Sketch	10^{-4}	1×10^{-5}	2×10^{-5}	3×10^{-2}	8×10^{-1}	7×10^{-5}	7×10^{-5}	5×10^{-2}	2×10^{0}
	10^{-6}	2×10^{-10}	3×10^{-10}	1×10^{-4}	4×10^{-3}	4×10^{-12}	2×10^{-11}	1×10^{-5}	1×10^{-3}
	10^{-8}	6×10^{-16}	2×10^{-15}	1×10^{-7}	9×10^{-6}	2×10^{-15}	1×10^{-14}	5×10^{-7}	3×10^{-5}

the path flow bound can be considerably larger than the true path flow error in certain cases. Although the path flow bound may not appear to be sufficiently tight in some cases, it is small relative to the equilibrium path flows and may therefore be sufficient for most practical purposes. For instance, across all four networks, an RGAP of 10^{-4} yields a path flow bound of at most 0.8% of $\|\hat{\boldsymbol{h}}\|$ for $\theta=0.5$ and 4% of $\|\hat{\boldsymbol{h}}\|$ for $\theta=1.5$. We also find that at the same RGAP both the true errors and the error bounds are generally larger for $\theta=1.5$ than for $\theta=0.5$. This behavior is expected, as increasing θ reduces the curvature of the objective contributed by the entropy term, giving a larger upper bound on the path flow error for a fixed objective difference.

We now turn to the link flow and link cost bounds in Theorem 3 and Corollary 1. Since both the link flow and link cost error bounds are derived for each individual link, it is not practical to analyze their tightness link by link. Rather, we focus on the norm of their errors. For instance, Theorem 3 gives $|a_l-\hat{a}_l|\le m_a^l$ for each link $l\in\mathbb{L}$, which we convert into a bound on the norm, $\|\boldsymbol{a}-\hat{\boldsymbol{a}}\|\le\|(m_a^l)_{l\in\mathbb{L}}\|$, where $(m_a^l)_{l\in\mathbb{L}}$ refers to the vector of link flow bounds over all links. Similarly, the link cost error bounds in Corollary 1 give $\|\mathcal{T}(\boldsymbol{a})-\mathcal{T}(\hat{\boldsymbol{a}})\|\le\|(L_a^l m_a^l)_{l\in\mathbb{L}}\|$. We look at these errors in relative terms, using the formulae shown in Table 4.

Table 6 reports these relative errors and error bounds at the same three RGAP values and two values of θ. Both bounds remain valid in all cases, and they shrink quickly as the RGAP decreases. For instance, at $\theta=0.5$, the bounds guarantee that the link flows and link costs are within 2% of their equilibrium values at an RGAP of 10^{-4} in all four networks. At an RGAP of 10^{-6}, the link flows and link costs are guaranteed to be within 0.2% of their equilibrium values for both $\theta=0.5$ and $\theta=1.5$ in all four networks.

Table 6: Link flow and link cost errors and bounds, in percent of the equilibrium value.

			$\theta = 0.5$		$\theta = 1.5$	
Network		**RGAP**	**True error %**	**Error bound %**	**True error %**	**Error bound %**
Sioux Falls	Link flow	10^{-4}	3×10^{-4}	5×10^{-1}	7×10^{-3}	1×10^{1}
		10^{-6}	6×10^{-6}	2×10^{-2}	3×10^{-5}	2×10^{-1}
		10^{-8}	9×10^{-8}	8×10^{-4}	2×10^{-7}	3×10^{-2}
	Link cost	10^{-4}	4×10^{-4}	6×10^{-1}	5×10^{-3}	2×10^{1}
		10^{-6}	1×10^{-5}	2×10^{-2}	1×10^{-4}	2×10^{-1}
		10^{-8}	2×10^{-7}	9×10^{-4}	6×10^{-7}	2×10^{-2}
Anaheim	Link flow	10^{-4}	8×10^{-3}	3×10^{-1}	1×10^{-2}	4×10^{-1}
		10^{-6}	4×10^{-5}	2×10^{-3}	8×10^{-5}	4×10^{-3}
		10^{-8}	6×10^{-7}	2×10^{-5}	1×10^{-6}	6×10^{-5}
	Link cost	10^{-4}	2×10^{-2}	1×10^{-1}	1×10^{-2}	9×10^{-2}
		10^{-6}	4×10^{-5}	8×10^{-4}	3×10^{-5}	9×10^{-4}
		10^{-8}	1×10^{-6}	8×10^{-6}	6×10^{-7}	1×10^{-5}
Winnipeg	Link flow	10^{-4}	6×10^{-4}	1×10^{-1}	1×10^{-2}	1×10^{0}
		10^{-6}	2×10^{-5}	1×10^{-3}	3×10^{-4}	2×10^{-1}
		10^{-8}	2×10^{-6}	4×10^{-5}	1×10^{-6}	8×10^{-3}
	Link cost	10^{-4}	9×10^{-4}	6×10^{-2}	5×10^{-3}	4×10^{-1}
		10^{-6}	1×10^{-5}	5×10^{-4}	3×10^{-4}	7×10^{-2}
		10^{-8}	1×10^{-6}	2×10^{-5}	6×10^{-7}	3×10^{-3}
Chicago Sketch	Link flow	10^{-4}	5×10^{-3}	2×10^{0}	7×10^{-3}	7×10^{0}
		10^{-6}	8×10^{-5}	1×10^{-2}	1×10^{-5}	3×10^{-3}
		10^{-8}	1×10^{-7}	3×10^{-5}	2×10^{-7}	9×10^{-5}
	Link cost	10^{-4}	3×10^{-2}	2×10^{0}	1×10^{-2}	3×10^{1}
		10^{-6}	1×10^{-4}	5×10^{-3}	3×10^{-5}	9×10^{-4}
		10^{-8}	4×10^{-7}	1×10^{-5}	1×10^{-6}	3×10^{-5}

Table 7: TSTT errors and bounds, in percent of the equilibrium TSTT.

		$\theta = 0.5$		$\theta = 1.5$	
Network	**RGAP**	**True error %**	**Error bound %**	**True error %**	**Error bound %**
Sioux Falls	10^{-4}	4×10^{-4}	8×10^{-1}	7×10^{-3}	2×10^{1}
	10^{-6}	4×10^{-6}	3×10^{-2}	1×10^{-5}	5×10^{-1}
	10^{-8}	2×10^{-8}	1×10^{-3}	2×10^{-7}	6×10^{-2}
Anaheim	10^{-4}	4×10^{-3}	3×10^{-2}	3×10^{-3}	3×10^{-2}
	10^{-6}	8×10^{-6}	2×10^{-4}	7×10^{-6}	3×10^{-4}
	10^{-8}	8×10^{-8}	2×10^{-6}	7×10^{-8}	5×10^{-6}
Winnipeg	10^{-4}	7×10^{-5}	2×10^{-2}	3×10^{-3}	2×10^{-1}
	10^{-6}	2×10^{-6}	1×10^{-4}	2×10^{-5}	3×10^{-2}
	10^{-8}	3×10^{-7}	6×10^{-6}	2×10^{-7}	1×10^{-3}
Chicago Sketch	10^{-4}	2×10^{-3}	3×10^{-1}	3×10^{-3}	2×10^{0}
	10^{-6}	6×10^{-7}	1×10^{-3}	4×10^{-7}	3×10^{-4}
	10^{-8}	5×10^{-9}	3×10^{-6}	1×10^{-8}	8×10^{-6}

Finally, we analyze the TSTT bound in Theorem 4. Table 7 reports the true error in TSTT and its error bound at the same three RGAP values and two values of θ using the formulae shown in Table 4. Although the error bound often exceeds the true error significantly, it decreases rapidly as the RGAP decreases. At an RGAP of 10^{-6}, the bound is within 0.05% of the equilibrium TSTT for $\theta = 0.5$ and within 0.5% of equilibrium TSTT for $\theta = 1.5$ across all four networks.

Figure 3 shows the true errors and the error bounds for link flows, link costs, and TSTT as the iterations progress in the Chicago Sketch network, each relative to its equilibrium value. A horizontal dotted line marks a relative error of 1%. The figure also shows the link flow bound $\tilde{m}_a^l$ of Lemma 2 and the link cost and TSTT bounds obtained from it, that is, the bounds before the strengthening in Theorem 3. In the first few iterations, all three bounds are several orders of magnitude larger than the true errors and are too loose to be useful. However, once the RGAP falls to around 10^{-4}, the bounds decrease at about the same rate as the true errors. As a result, the bounds fall within 1% of the equilibrium values only a few iterations after the true errors do. The true errors, however, cannot be computed while running the traffic assignment unless the equilibrium is known beforehand. The bounds, on the other hand, are available at every iteration and guarantee a relative error of at most 1% for all three metrics once the RGAP falls between 10^{-4} and 10^{-6}.

We strengthen the bounds on TSTT for convex link performance functions in Corollary 2. To test the strengthening, we plot the current TSTT, the interval given by Theorem 4, and the convex lower bound from Corollary 2, all as a percentage difference from the equilibrium TSTT. Theorem 4 gives the interval

$$\begin{aligned} \zeta(\hat{\boldsymbol{a}}) &\geq \zeta(\boldsymbol{a}) - Q_a\sqrt{\frac{\theta\gamma(\boldsymbol{h})}{2}} - \frac{1}{2}\sum_{l\in\mathbb{L}} K_a^l (m_a^l)^2, \\ \zeta(\hat{\boldsymbol{a}}) &\leq \zeta(\boldsymbol{a}) + Q_a\sqrt{\frac{\theta\gamma(\boldsymbol{h})}{2}} + \frac{1}{2}\sum_{l\in\mathbb{L}} K_a^l (m_a^l)^2, \end{aligned} \tag{58}$$

and Corollary 2 gives the convex lower bound $\zeta(\boldsymbol{a}) - Q_a\sqrt{\theta\gamma(\boldsymbol{h})/2}$. Figure 4 shows these values on a symmetric log scale for the Sioux Falls, Winnipeg, and Chicago Sketch networks. We also mark differences of $\pm 1\%$ from the equilibrium TSTT. We find that the strengthened lower bound for convex link performance functions is orders of magnitude tighter than the lower end of the interval from Theorem 4, particularly in the early iterations. This is expected, as Corollary 2 discards the second-order term in Equation (51), which dominates the bound when the path flows are far from equilibrium. The strengthened lower bound

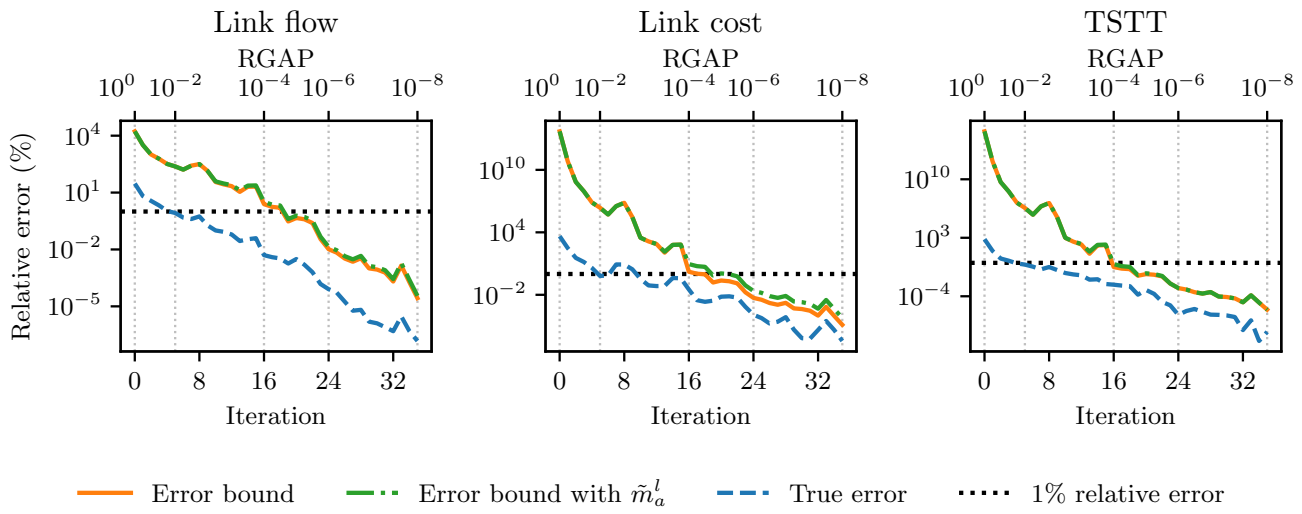


Figure 3: Relative errors, error bounds, and error bounds based on Lemma 2 versus iteration for the Chicago Sketch network (log scale, $\theta = 0.5$).

is therefore particularly useful in applications that only need to know how small the equilibrium TSTT can be, as we will see in the network design problem in Section 6.2.

6.2 Application to network design

To demonstrate how our bounds can be used in practice, we present a network design example in which a planning agency decides where to add road capacity on the Chicago Sketch network. We define a "design" as a set of links $\mathbb{U} \subseteq \mathbb{L}$ whose capacity is doubled. The agency has a budget of B miles, so a design is feasible only if $\sum_{l\in\mathbb{U}} b_l \le B$, where b_l is the length of link l in miles. We denote the current capacity of a link l by u_l. Assuming that adding a lane costs \$6 million per lane-mile and that the capacity of a lane is 1,900 vehicles per hour, the cost of expanding the link becomes $\$6 \times 10^6/1{,}900 \approx \$3{,}158$ per mile for each unit of added capacity. Since doubling the capacity of link l adds u_l units of capacity, the construction cost of design $\mathbb{U}$ is $C(\mathbb{U}) = 3{,}158 \sum_{l\in\mathbb{U}} b_l\, u_l$ dollars.

The objective is to find the design that minimizes the weighted sum of the construction cost and the value of TSTT,

$$Z(\mathbb{U}) = V\, \zeta\big(\hat{\boldsymbol{a}}(\mathbb{U})\big) + C(\mathbb{U}), \tag{59}$$

where $\hat{\boldsymbol{a}}(\mathbb{U})$ is the SUE link flow vector of design $\mathbb{U}$ and V converts TSTT into dollars. We assume that the input demand represents one hour of travel demand and that travel times are measured in minutes. We further assume that we are interested only in evaluating these designs over 500 peak hours per year for 30 years. With a 4% annual discount rate and a value of time of \$20 per vehicle-hour, a reasonable estimate of V is $1/60 \times 20 \times 500 \times \frac{1-(1.04)^{-30}}{0.04} \approx \$2{,}882$ per vehicle-minute. We call $Z(\emptyset) - Z(\mathbb{U})$ the net benefit of $\mathbb{U}$.

Evaluating Z for each design requires solving a traffic assignment, and the number of possible designs grows combinatorially with the budget. For instance, if the budget allowed only five links of equal length, an exhaustive search would have to solve a traffic assignment for every subset of one to five candidate links, that is, $\sum_{k=1}^{5} \binom{N}{k}$ traffic assignments for N candidate links. For the Chicago Sketch network, we take all 2,176 links other than the centroid connectors as candidate links. An exhaustive search on this set would require about 4×10^{14} traffic assignment runs.

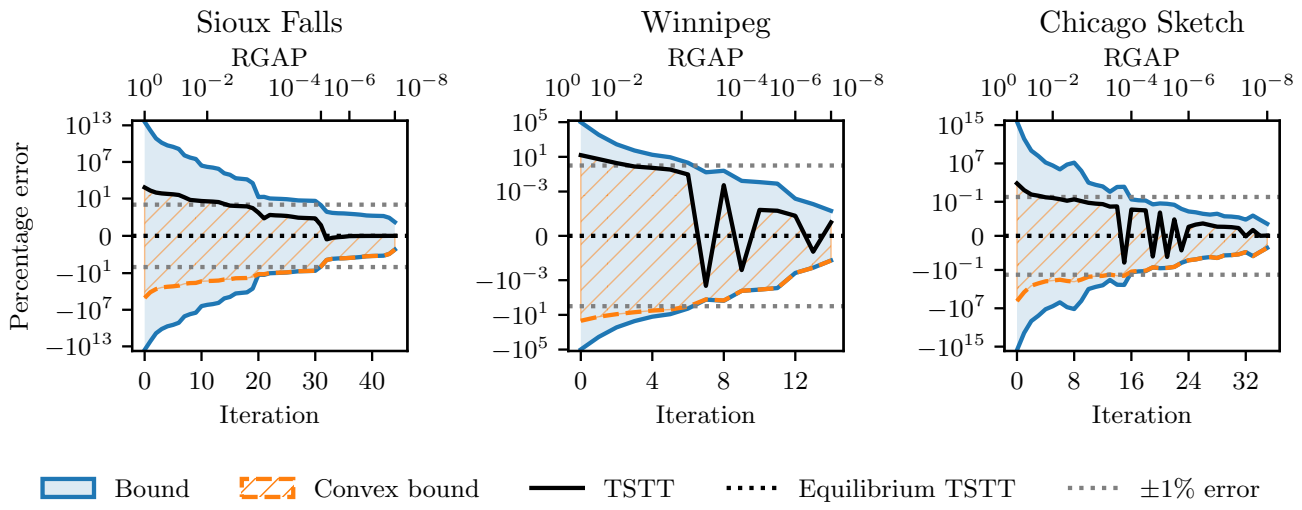


Figure 4: TSTT and its bounds versus iteration (symmetric log scale, $\theta = 0.5$).

The goal of this subsection is not to quantify how much computation time the bounds can save in network design, but to illustrate how they can be applied. Hence, even though many methods have been developed for network design problems, we use a simple first-improvement local search to find a locally optimal design. Starting from a random design within the budget, we consider every neighbor of the current design $\mathbb{U}$, that is, every design obtained by adding, dropping, or swapping one upgraded link while staying within budget, and try these neighbors in random order. The first neighbor $\mathbb{U}'$ that improves the objective, $Z(\mathbb{U}') < Z(\mathbb{U})$, becomes the current design, and the neighbors of the new design are then tried in a new random order. The search stops when none of the neighbors improves the objective, yielding a local minimum, or when the time limit is reached.

Without using the bounds, every traffic assignment is solved to an RGAP of 10^{-12}. With bounds, we use Corollary 2 at any path flow vector $\boldsymbol{h}$ of the assignment for $\mathbb{U}'$ to obtain the lower bound $\underline{Z}(\mathbb{U}') = V\left(\zeta(\boldsymbol{a}) - Q_a\sqrt{\theta\gamma(\boldsymbol{h})/2}\right) + C(\mathbb{U}') \le Z(\mathbb{U}')$, and reject $\mathbb{U}'$ as soon as $\underline{Z}(\mathbb{U}') \ge Z(\mathbb{U})$. We refer to such a design as "pruned", since the design cannot improve on the current objective even if its traffic assignment is solved exactly. As shown in Algorithm 1, we check the lower bound on Z at a sequence of decreasing RGAP values, resuming the same assignment between checks, and solve a neighbor to an RGAP of 10^{-12} only if it is never pruned. We treat the TSTT at an RGAP of 10^{-12} as the equilibrium TSTT. In practice, a looser RGAP such as 10^{-4} is often used to save computation, but then a design may be accepted or rejected based on a TSTT that has not yet converged. However, when each design is solved to an RGAP of 10^{-12} without bounds, a design pruned using the bounds would also have been rejected without bounds. The local search with bounds and the search without bounds then make the same moves in the same order. The bounds only decide how far each assignment must be solved before we can be certain whether to reject the design, which allows a fair comparison of the run times of the two searches.

We ran the local search on the Chicago Sketch network with $\theta = 0.5$, a budget of $B = 40$ miles, and a time limit of one hour, once with and once without the bounds, using `FP-BB1` and the same random seed. Without bounds, a total of 237 designs were evaluated in one hour, while using the bounds, 453 designs were evaluated in the same computational budget. Of the 453 designs evaluated with bounds, 433 (96%) were pruned before their traffic assignment reached an RGAP of 10^{-12}, and apart from the start design, every design solved to 10^{-12} led to an improvement in objective. The right panel of Figure 5 shows the RGAP at which each design was pruned and its change in Z from the current design. Among the pruned designs, 88 designs were pruned at an RGAP of 10^{-4}, 324 at 10^{-5}, 18 at 10^{-6}, 2 at 10^{-7}, and

Algorithm 1 Local search with bound-based pruning

Draw a random design $\mathbb{U}$ within the budget and compute $Z(\mathbb{U})$ at an RGAP of 10^{-12}.
for each neighbor $\mathbb{U}'$ of $\mathbb{U}$, in random order, until the time limit is reached **do**
    for each RGAP threshold in $\{10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}\}$ **do**
        Run (or resume) the traffic assignment of $\mathbb{U}'$ until the RGAP falls below this threshold.
        if $\underline{Z}(\mathbb{U}') \geq Z(\mathbb{U})$ **then** prune $\mathbb{U}'$ and go to the next neighbor.
    end for
    Resume until an RGAP of 10^{-12}.
    if $Z(\mathbb{U}') < Z(\mathbb{U})$ **then** $\mathbb{U} \leftarrow \mathbb{U}'$ and restart the loop over neighbors.
end for
return $\mathbb{U}$.

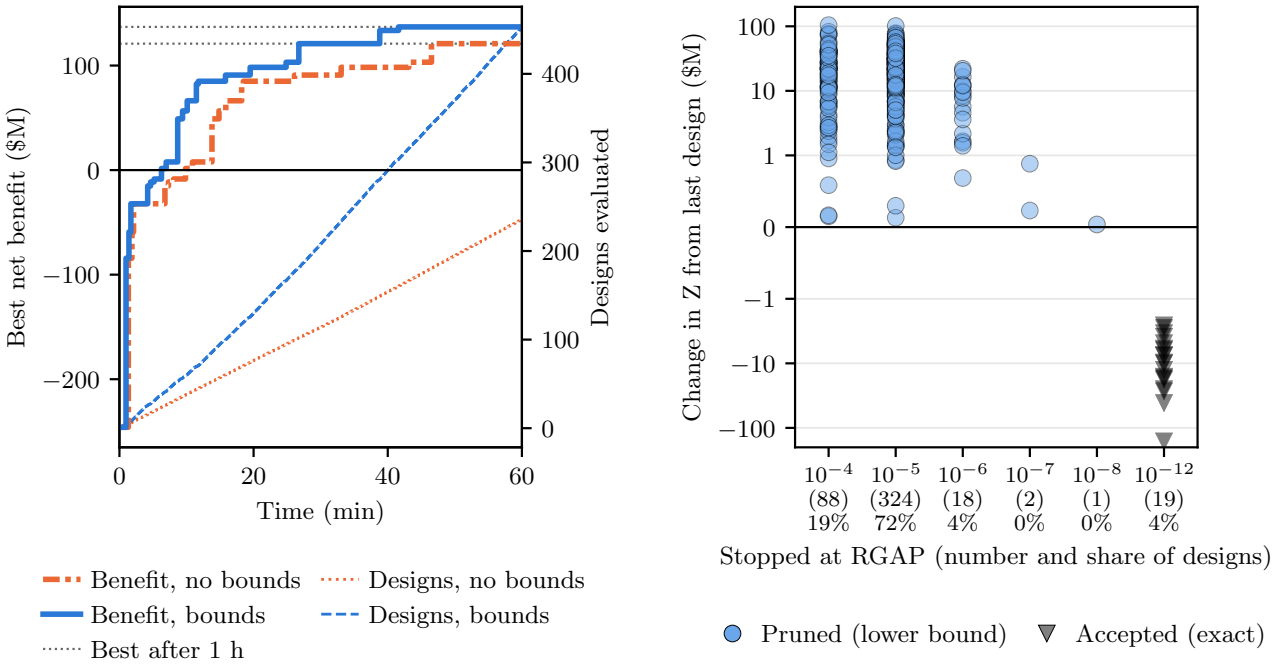


Figure 5: Local search for capacity design on the Chicago Sketch network.

1 at 10^{-8}. Hence, in this network, no single fixed tolerance served all designs. An RGAP of 10^{-4} was not sufficient to certify the termination for 364 designs, while an RGAP of 10^{-12} was never necessary except for improving designs.

As a consequence, the bounded search also saved time. It completed all 237 evaluations that the search without bounds made in one hour in only 2,003 seconds, a 45% reduction in run time. With the time saved, it went on to find a better design. As shown in the left panel of Figure 5, its best design after one hour had a net benefit of $137.0 million, compared with $120.9 million without bounds, an improvement of 13%.

7 Conclusion

In this article, we present error bounds on the path flows, link flows, link costs, and TSTT of any feasible solution to the stochastic TAP relative to the equilibrium solution. We start by deriving an upper bound on the error in the SUE objective using a partial linearization of Fisk's objective function for logit-SUE.

We then show that the strong convexity of the SUE objective allows the path flow errors to be bounded above in terms of the error in the SUE objective. Using this bound, we derive an upper bound on the distance of any feasible path flow solution to the stochastic TAP from the SUE solution.

Error bounds on link flows, link costs, and TSTT follow directly from the path flow bound. To derive an error bound on the link flows, we first aggregate the path flow errors across all paths using a link and then tighten the bound using demand conservation. Thereafter, we show that the error bound on link flows can be further tightened using the slope of the link performance functions. We then bound the link costs using Lipschitz continuity, and TSTT using the path flow and link flow bounds. Finally, we show that the lower bound on TSTT can be made significantly tighter when the link performance functions are convex.

Results on city-level networks show that the bounds can guarantee how far a current solution is from the equilibrium solution in terms of the metric under consideration, which would allow planners to use them as convergence criteria for that metric. At an RGAP of 10^{-6}, the bounds guarantee that the path flows, link flows, link costs, and TSTT are all within 0.1% of their equilibrium values for $\theta = 0.5$, and within 1% for $\theta = 1.5$. Furthermore, we show that the bounds converge to zero at least as fast as the path flows converge to equilibrium, regardless of the algorithm used. We also find that computing all the bounds takes less time than a single iteration of `FP-BB1`, which is small compared with the entire TAP procedure.

To show the applicability of the bounds, we present a network design case study, in which we model a capacity design problem that minimizes a weighted sum of the TSTT and the construction cost. With the bounds, 96% of the candidate designs were rejected before their traffic assignment was solved to a high accuracy. This led to a 45% reduction in computation time and, within the same one-hour time limit, to a design with a 13% higher net benefit. We note that the time the bounds can save is itself limited, since a pruned design still needs its traffic assignment to reach an RGAP of 10^{-4}. In this example, since `FP-BB1` converges linearly, reaching an RGAP of 10^{-4} took on average 40% of the time needed to reach an RGAP of 10^{-12}. Even if every rejected design had been pruned at 10^{-4}, the computation time would have been reduced by only 57%, since the start design and every accepted design still had to be solved to an RGAP of 10^{-12}. The bounded search therefore already achieved most of the possible improvement. How best to use the bounds within a given algorithm warrants a separate analysis outside the scope of this paper.

Although this article focuses on logit-SUE with fixed acyclic path sets, many of our results can be extended to several variants of the TAP and to different path sets. For instance, relaxing the path sets to include paths with cycles is straightforward. In that case, only the matrix $\boldsymbol{D}$ changes, as its entries are no longer binary but count the number of times a path traverses a link. The objective, path flow, and convergence results do not depend on $\boldsymbol{D}$ being binary, so the only part of our derivation that needs to change is the link flow bound in Lemma 2. The idea that the link flow error is a sum of path flow errors still applies, except that the demand conservation step in Equation (37), where we used the fact that each path uses a link at most once, needs to be updated, as does the definition of $\mathbb{W}_l$. This changes only the leading constants in Lemma 2. All other bounds then follow directly.

Even for other variants of logit-SUE, and in fact even for deterministic user equilibrium, many of the ideas used in this study would carry over. The main idea of linearizing the objective and connecting the flows to the linearization stays the same in many variants. Once the path flows and link flows are bounded, the arguments used to bound link costs and TSTT follow naturally, as they do not use any property specific to the stochastic TAP. Deriving and testing the error bounds for these different TAP variants therefore remains a promising direction for future work, although outside the scope of the current study.

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Appendix A. Minimizer of the partially linearized SUE objective

For completeness, we prove the property of ψ_h used in Theorem 1. The result is standard; see Leurent (1997).

Lemma 5. *Fix a feasible path flow vector $\boldsymbol{h}$ and let*

$$\psi_h(\tilde{\boldsymbol{h}}) = \beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T(\tilde{\boldsymbol{h}} - \boldsymbol{h}) + \eta(\tilde{\boldsymbol{h}}), \tag{60}$$

as defined in Equation (13). Then ψ_h has a unique minimizer over $\mathbb{H}$, that minimizer is strictly positive, and it equals $\mathcal{L}(\boldsymbol{h})$.

Proof. We know η is strictly convex on $\mathbb{H}$. ψ_h adds a linear function of $\tilde{\boldsymbol{h}}$ to η, so ψ_h is continuous and strictly convex on $\mathbb{H}$. The set $\mathbb{H}$ is nonempty and compact, so a minimizer exists and is unique.

It remains to show that this minimizer is $\mathcal{L}(\boldsymbol{h})$. Since $\mathcal{L}(\boldsymbol{h}) > \mathbf{0}$, $\forall \boldsymbol{h} \in \mathbb{H}$ (by Equation (5)), ψ_h is differentiable at $\mathcal{L}(\boldsymbol{h})$, and for every $i \in \mathbb{P}$,

$$\begin{aligned}\frac{\partial \psi_h}{\partial \tilde{h}_i}\big(\mathcal{L}(\boldsymbol{h})\big) &= \mathcal{C}_i(\boldsymbol{h}) + \frac{1}{\theta}\big[1 + \ln \mathcal{L}_i(\boldsymbol{h})\big] \\ &= \frac{1}{\theta}\Big[1 + \ln d_{w(i)} - \ln \sum_{j \in \mathbb{P}_{w(i)}} \exp[-\theta \mathcal{C}_j(\boldsymbol{h})]\Big],\end{aligned} \tag{61}$$

where the second step substitutes Equation (4), and the terms $\mathcal{C}_i(\boldsymbol{h})$ cancel. The right-hand side is the same for all paths in an OD pair. Since every $\tilde{\boldsymbol{h}} \in \mathbb{H}$ satisfies $\sum_{i \in \mathbb{P}_w} \big(\tilde{h}_i - \mathcal{L}_i(\boldsymbol{h})\big) = 0$ for every $w \in \mathbb{W}$, it follows that $\nabla \psi_h(\mathcal{L}(\boldsymbol{h}))^T(\tilde{\boldsymbol{h}} - \mathcal{L}(\boldsymbol{h})) = 0$. By convexity of ψ_h, for every $\tilde{\boldsymbol{h}} \in \mathbb{H}$,

$$\begin{aligned}\psi_h(\tilde{\boldsymbol{h}}) &\geq \psi_h(\mathcal{L}(\boldsymbol{h})) \\ &\quad + \nabla \psi_h(\mathcal{L}(\boldsymbol{h}))^T\big(\tilde{\boldsymbol{h}} - \mathcal{L}(\boldsymbol{h})\big) \\ &= \psi_h(\mathcal{L}(\boldsymbol{h})),\end{aligned} \tag{62}$$

so $\mathcal{L}(\boldsymbol{h})$ is the minimizer. □

Appendix B. Proof of lemmas used in the TSTT bound

B.1 Proof of Lemma 3

We want to show that for any feasible path flow vector $\boldsymbol{h}$, if we define

$$Q_a := \sqrt{\sum_{w \in \mathbb{W}} d_w \left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i\right)^2}, \tag{63}$$

then the first-order term in Equation (44) given by Equation (45) satisfies

$$|\boldsymbol{q}^T \boldsymbol{v}| \leq Q_a \sqrt{\frac{\theta \gamma(\boldsymbol{h})}{2}}. \tag{64}$$

Proof. We first bound the contribution to $\boldsymbol{q}^T\boldsymbol{v}$ from one OD pair. Since $\sum_{i \in \mathbb{P}_w} v_i = 0$, the positive and negative components of $\boldsymbol{v}^w$ each have total magnitude $\|\boldsymbol{v}^w\|_1/2$. Separating them,

$$\begin{aligned}\left|\sum_{i \in \mathbb{P}_w} q_i v_i\right| &= \left|\sum_{\substack{i \in \mathbb{P}_w \\ v_i > 0}} q_i v_i - \sum_{\substack{i \in \mathbb{P}_w \\ v_i < 0}} q_i |v_i|\right| \\ &\leq \frac{1}{2}\left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i\right) \|\boldsymbol{v}^w\|_1.\end{aligned} \tag{65}$$

Summing Equation (65) over OD pairs, and multiplying and dividing each term by $\sqrt{d_w}$, gives

$$\begin{aligned} |\boldsymbol{q}^T \boldsymbol{v}| &\leq \frac{1}{2} \sum_{w \in \mathbb{W}} \left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i \right) \|\boldsymbol{v}^w\|_1 \\ &= \frac{1}{2} \sum_{w \in \mathbb{W}} \left[\sqrt{d_w} \left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i \right) \right] \frac{\|\boldsymbol{v}^w\|_1}{\sqrt{d_w}}. \end{aligned} \tag{66}$$

Finally, applying the Cauchy-Schwarz inequality to this sum gives

$$\begin{aligned} &|\boldsymbol{q}^T \boldsymbol{v}| \\ &\leq \frac{1}{2} \sqrt{\sum_{w \in \mathbb{W}} d_w \left(\max_{i \in \mathbb{P}_w} q_i - \min_{i \in \mathbb{P}_w} q_i \right)^2} \sqrt{\sum_{w \in \mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}}. \end{aligned} \tag{67}$$

The first square root equals Q_a. By Equation (30) in Theorem 2, $\sum_{w \in \mathbb{W}} \|\boldsymbol{v}^w\|_1^2 / d_w \leq 2\theta\gamma(\boldsymbol{h})$. Substituting this into Equation (67) proves the result. □

B.2 Proof of Lemma 4

We want to show that, for every feasible path flow vector $\boldsymbol{h}$, if we define

$$K_a^l := \max_{\tilde{a}_l \in \mathbb{A}_a^l} \left| 2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right|, \qquad l \in \mathbb{L}, \tag{68}$$

then the second-order term in Equation (44) given by Equation (46) satisfies

$$\begin{aligned} &\left| \frac{1}{2} \sum_{l \in \mathbb{L}} \left[2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right] (\hat{a}_l - a_l)^2 \right| \\ &\qquad \leq \frac{1}{2} \sum_{l \in \mathbb{L}} K_a^l (m_a^l)^2, \end{aligned} \tag{69}$$

where m_a^l is given by Theorem 3.

Proof. Theorem 3 gives $|\hat{a}_l - a_l| \leq m_a^l$, so both a_l and $\hat{a}_l$ belong to $\mathbb{A}_a^l$. Since $\tilde{a}_l$ lies between these two flows, $\tilde{a}_l \in \mathbb{A}_a^l$. The definition of K_a^l therefore gives

$$\left| 2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right| \leq K_a^l. \tag{70}$$

Applying the triangle inequality then proves the result:

$$\begin{aligned} &\left| \frac{1}{2} \sum_{l \in \mathbb{L}} \left[2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right] (\hat{a}_l - a_l)^2 \right| \\ &\leq \frac{1}{2} \sum_{l \in \mathbb{L}} \left| 2\tau_l'(\tilde{a}_l) + \tilde{a}_l \tau_l''(\tilde{a}_l) \right| (\hat{a}_l - a_l)^2 \\ &\leq \frac{1}{2} \sum_{l \in \mathbb{L}} K_a^l (\hat{a}_l - a_l)^2 \leq \frac{1}{2} \sum_{l \in \mathbb{L}} K_a^l (m_a^l)^2. \quad \square \end{aligned} \tag{71}$$

Appendix C. Tightening of the link flow bound

We now prove Theorem 3. For a link $l \in \mathbb{L}$, we recall that

$$R_a^l := \min\Big\{\tau_l'(\tilde{a}_l) : \ \tilde{a}_l \geq 0, \ |\tilde{a}_l - a_l| \leq \tilde{m}_a^l\Big\}, \ S_l := \sum_{w \in \mathbb{W}_l} d_w, \tag{72}$$

and we want to show that, for any feasible path flow vector $\boldsymbol{h}$,

$$|a_l - \hat{a}_l| \leq m_a^l := \frac{\tilde{m}_a^l}{\sqrt{1 + \theta S_l R_a^l/4}}. \tag{73}$$

The proof follows Lemma 1, Theorem 2, and Lemma 2. In the proof of Lemma 1, we used only the fact that the second derivative of the cost component β is non-negative and discarded it. Here, we will bound this second derivative from below using R_a^l and track this bound through the proofs of Theorem 2 and Lemma 2.

Proof. Fix a feasible path flow vector $\boldsymbol{h}$. As in the proof of Lemma 1, we write $\boldsymbol{v} := \boldsymbol{h} - \hat{\boldsymbol{h}}$ and $\hbar(t) := \hat{\boldsymbol{h}} + t\boldsymbol{v}$ for $0 \leq t \leq 1$. Then the link flows along this segment for each link $j \in \mathbb{L}$ are

$$\boldsymbol{a}_j(t) = (1-t)\hat{a}_j + t a_j, \qquad \frac{d}{dt}\boldsymbol{a}_j(t) = a_j - \hat{a}_j. \tag{74}$$

Along this segment, $\beta(\hbar(t)) = \sum_{j \in \mathbb{L}} \int_0^{\boldsymbol{a}_j(t)} \tau_j(x)\,dx$. By the fundamental theorem of calculus and the chain rule, together with Equation (74),

$$\frac{d^2}{dt^2}\beta(\hbar(t)) = \sum_{j \in \mathbb{L}} \tau_j'\big(\boldsymbol{a}_j(t)\big)(a_j - \hat{a}_j)^2. \tag{75}$$

In the proof of Lemma 1, we only used the fact that this second derivative is non-negative. But now we keep it and bound it from below. Every term in the sum in the right-hand side of Equation (75) is non-negative because each τ_j is non-decreasing, so that $\tau_j' \geq 0$. Keeping only the term of link l gives

$$\frac{d^2}{dt^2}\beta(\hbar(t)) \geq \tau_l'\big(\boldsymbol{a}_l(t)\big)(a_l - \hat{a}_l)^2 \geq R_a^l(a_l - \hat{a}_l)^2. \tag{76}$$

The last inequality follows from the definition of R_a^l in Equation (72) as both a_l and $\hat{a}_l$ lie in the interval $\{x \geq 0 : \ |x - a_l| \leq \tilde{m}_a^l\}$, so $\boldsymbol{a}_l(t)$, which lies between them, lies in this interval as well.

We already showed in the proof of Lemma 1 that Equation (25) and Inequality (26) give $\frac{d^2}{dt^2}\eta(\hbar(t)) \geq \frac{1}{\theta}\sum_{w \in \mathbb{W}} \|\boldsymbol{v}^w\|_1^2/d_w$ for every $t \in [0,1)$. Adding this to Inequality (76), for every $t \in [0,1)$,

$$\frac{d^2}{dt^2}\phi(\hbar(t)) \geq R_a^l(a_l - \hat{a}_l)^2 + \frac{1}{\theta}\sum_{w \in \mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}. \tag{77}$$

We now apply Taylor's theorem as we did in Equation (28) in the proof of Lemma 1 to obtain a bound on $\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}})$. There, we integrated up to $s < 1$ and then let s approach one using the continuity of ϕ on $\mathbb{H}$. The same limiting argument applies here, so for brevity we integrate directly up to one. Since the first derivative of $\phi(\hbar(t))$ is zero at $t = 0$ by Equation (23),

$$\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) = \int_0^1 (1-t)\frac{d^2}{dt^2}\phi(\hbar(t))\,dt. \tag{78}$$

Since $1-t \geq 0$ for all $t \in [0,1]$, we can replace the second derivative $\frac{d^2}{dt^2}\phi(\hbar(t))$ by its lower bound in Inequality (77). This lower bound does not depend on t, so using $\int_0^1 (1-t)\,dt = 1/2$,

$$\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) \geq \frac{R_a^l}{2}(a_l - \hat{a}_l)^2 + \frac{1}{2\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}. \tag{79}$$

We note here that the bound in Inequality (79) is exactly the bound in Lemma 1 but with the bound on the second derivative of the SUE objective's cost term tracked. In the proof of Theorem 2, the next step was to combine Lemma 1 with Theorem 1 to "sandwich" the error in the SUE objective. We will now do the same with the bound in (79). Combining this with Theorem 1, which bounds the error in the SUE objective from above by $\gamma(\boldsymbol{h})$,

$$\frac{1}{2\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w} + \frac{R_a^l}{2}(a_l - \hat{a}_l)^2 \leq \gamma(\boldsymbol{h}). \tag{80}$$

In Theorem 2, we only had the first term on the left. The second term is what tightens the link flow bound.

The final step is to write the path flow errors in terms of link flow errors, which completes the proof. To do this, we will follow the same argument as in Lemma 2 used to bound the link flows. If $S_l = 0$, then $\mathbb{W}_l$ is empty, $a_l = \hat{a}_l$, and the result holds trivially. So we assume $S_l > 0$. We showed in Equation (38) that $|a_l - \hat{a}_l| \leq \frac{1}{2}\sqrt{S_l \sum_{w\in\mathbb{W}_l} \|\boldsymbol{v}^w\|_1^2/d_w}$. Squaring the terms, enlarging the sum to all OD pairs since every term is non-negative, cross-multiplying, and multiplying by $1/(2\theta)$ on both sides,

$$\frac{2}{\theta S_l}(a_l - \hat{a}_l)^2 \leq \frac{1}{2\theta}\sum_{w\in\mathbb{W}} \frac{\|\boldsymbol{v}^w\|_1^2}{d_w}. \tag{81}$$

Substituting Inequality (81) into Inequality (80) gives

$$\frac{2}{\theta S_l}(a_l - \hat{a}_l)^2 + \frac{R_a^l}{2}(a_l - \hat{a}_l)^2 \leq \gamma(\boldsymbol{h}). \tag{82}$$

Multiplying both sides of Inequality (82) by $\theta S_l/2 > 0$, and using $(\tilde{m}_a^l)^2 = \theta S_l \gamma(\boldsymbol{h})/2$ from Lemma 2,

$$\left(1 + \frac{\theta S_l R_a^l}{4}\right)(a_l - \hat{a}_l)^2 \leq (\tilde{m}_a^l)^2. \tag{83}$$

Since $R_a^l \geq 0$, the factor $1 + \theta S_l R_a^l/4$ is at least one. Dividing by it and taking square roots gives the upper bound on link flows in Inequality (73). □

Appendix D. Proof of convergence theorems and corollaries

D.1 Proof of Theorem 5

We want to prove that, as $\boldsymbol{h}^{(k)} \to \hat{\boldsymbol{h}}$,

$$\begin{aligned} \gamma(\boldsymbol{h}^{(k)}) &= O\left(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|^2\right), \\ m_{h^{(k)}} &= O\left(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|\right). \end{aligned} \tag{84}$$

Proof. We recall that $\gamma(\boldsymbol{h}^{(k)}) = \psi_{h^{(k)}}(\boldsymbol{h}^{(k)}) - \psi_{h^{(k)}}(\mathcal{L}(\boldsymbol{h}^{(k)}))$ is the difference in $\psi_{h^{(k)}}$ between the current flows and the target flows. To bound this difference, we apply Taylor's theorem to $\psi_{h^{(k)}}$ along the segment joining the current and target flows,

$$g(t) := \mathcal{L}(\boldsymbol{h}^{(k)}) + t\big(\boldsymbol{h}^{(k)} - \mathcal{L}(\boldsymbol{h}^{(k)})\big), \tag{85}$$

$0 \le t \le 1$. Since $\mathcal{L}(\boldsymbol{h}^{(k)}) > \mathbf{0}$, every $\boldsymbol{g}(t)$ with $t \in [0,1)$ is strictly positive, so $\psi_{h^{(k)}}$ is twice differentiable along this segment. Taylor's theorem applied to $\psi_{h^{(k)}}$ up to the first order (with second-order remainder) along this segment gives some $0 \le \tilde{t} < 1$ such that

$$\begin{aligned}\psi_{h^{(k)}}(\boldsymbol{g}(1)) = \psi_{h^{(k)}}(\boldsymbol{g}(0)) + \frac{d}{dt}\psi_{h^{(k)}}(\boldsymbol{g}(t))\Big|_{t=0} \\ + \frac{1}{2}\frac{d^2}{dt^2}\psi_{h^{(k)}}(\boldsymbol{g}(t))\Big|_{t=\tilde{t}}.\end{aligned} \tag{86}$$

Moving the term $\psi_{h^{(k)}}(\boldsymbol{g}(0))$ to the left-hand side gives

$$\gamma(\boldsymbol{h}^{(k)}) = \frac{d}{dt}\psi_{h^{(k)}}(\boldsymbol{g}(t))\Big|_{t=0} + \frac{1}{2}\frac{d^2}{dt^2}\psi_{h^{(k)}}(\boldsymbol{g}(t))\Big|_{t=\tilde{t}}. \tag{87}$$

We now compute the two derivatives separately. For the first derivative, Lemma 5 shows that $\boldsymbol{g}(0) = \mathcal{L}(\boldsymbol{h}^{(k)})$ minimizes $\psi_{h^{(k)}}$ over $\mathbb{H}$ and is strictly positive in all components, so $\boldsymbol{g}(t)$ remains feasible in a small neighborhood of $t = 0$. Furthermore, since both $\boldsymbol{h}^{(k)}$ and $\mathcal{L}(\boldsymbol{h}^{(k)})$ satisfy the OD demands, moving in the direction $\boldsymbol{g}'(0) = \boldsymbol{h}^{(k)} - \mathcal{L}(\boldsymbol{h}^{(k)})$ preserves OD demand. Hence,

$$\frac{d}{dt}\psi_{h^{(k)}}(\boldsymbol{g}(t))\Big|_{t=0} = \nabla\psi_{h^{(k)}}(\boldsymbol{g}(0))^T\boldsymbol{g}'(0) = 0. \tag{88}$$

For the second derivative, we recall from Equation (13) that $\psi_{h^{(k)}}(\tilde{\boldsymbol{h}}) = \beta(\boldsymbol{h}^{(k)}) + \mathcal{C}(\boldsymbol{h}^{(k)})^T(\tilde{\boldsymbol{h}} - \boldsymbol{h}^{(k)}) + \eta(\tilde{\boldsymbol{h}})$. The path costs are held fixed, so the cost term is linear in $\tilde{\boldsymbol{h}}$ and only the entropy term η contributes to the second derivative. Using Equation (25) with $\boldsymbol{v}$ replaced by $\boldsymbol{h}^{(k)} - \mathcal{L}(\boldsymbol{h}^{(k)})$ and $\hbar(t)$ replaced by $\boldsymbol{g}(t)$ gives

$$\begin{aligned}\frac{d^2}{dt^2}\psi_{h^{(k)}}(\boldsymbol{g}(t)) &= \frac{d^2}{dt^2}\eta(\boldsymbol{g}(t)) \\ &= \frac{1}{\theta}\sum_{w\in\mathbb{W}}\sum_{i\in\mathbb{P}_w}\frac{\left(h_i^{(k)} - \mathcal{L}_i(\boldsymbol{h}^{(k)})\right)^2}{\boldsymbol{g}_i(t)}.\end{aligned} \tag{89}$$

To bound $\boldsymbol{g}_i(t)$ from below in Equation (89), we consider a ball of radius $\frac{1}{2}\min_j \hat{h}_j$ around $\hat{\boldsymbol{h}}$. Every point in this ball is at least $\frac{1}{2}\min_j \hat{h}_j$ in each component. Since $\boldsymbol{h}^{(k)} \to \hat{\boldsymbol{h}}$, and since $\mathcal{L}$ is continuous with $\mathcal{L}(\hat{\boldsymbol{h}}) = \hat{\boldsymbol{h}}$, both $\boldsymbol{h}^{(k)}$ and $\mathcal{L}(\boldsymbol{h}^{(k)})$ lie in this ball for all sufficiently large k. The ball is convex, so every $\boldsymbol{g}(t)$, being a convex combination of $\boldsymbol{h}^{(k)}$ and $\mathcal{L}(\boldsymbol{h}^{(k)})$, also lies in it. Hence, for every $t \in [0,1]$, $\boldsymbol{g}_i(t) \ge \frac{1}{2}\min_j \hat{h}_j$ for all i, and

$$\frac{d^2}{dt^2}\psi_{h^{(k)}}(\boldsymbol{g}(t)) \le \frac{2\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|^2}{\theta\min_j \hat{h}_j}. \tag{90}$$

Substituting the first derivative of $\psi_{h^{(k)}}(\boldsymbol{g}(t))$ from Equation (88) and the bound on the second derivative from Equation (90) into Equation (87),

$$\gamma(\boldsymbol{h}^{(k)}) \le \frac{\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|^2}{\theta\min_j \hat{h}_j}, \tag{91}$$

which proves the first part of the theorem. The path flow bound $m_{h^{(k)}} = \sqrt{\theta d_{\max}\gamma(\boldsymbol{h}^{(k)})}$ is a fixed multiple of $\sqrt{\gamma(\boldsymbol{h}^{(k)})}$, so taking square roots proves the second part. □

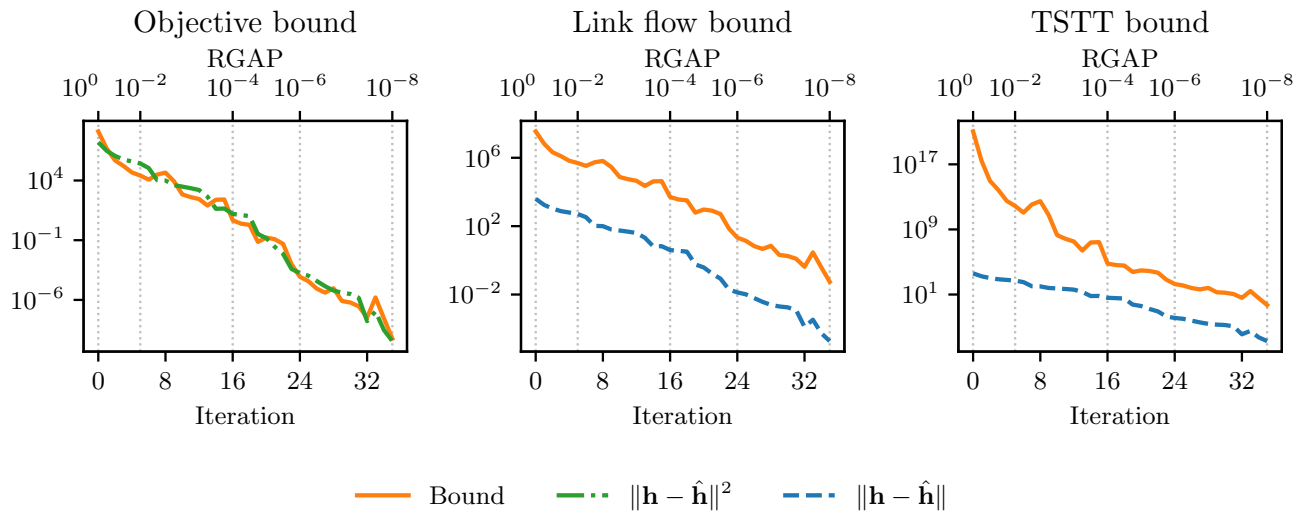


Figure 6: Objective, link flow, and TSTT bounds with $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^2$ or $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$ versus iteration for the Chicago Sketch network, from RGAP 10^{-4} on (log scale, $\theta = 0.5$).

D.2 Proof of Corollary 3

We want to prove that, as $\boldsymbol{h}^{(k)} \to \hat{\boldsymbol{h}}$, the link flow bounds m_a^l, the link cost bounds $L_a^l m_a^l$, and the TSTT bound in Theorem 4 are all $O(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|)$ and converge to zero.

Proof. The link flow bound m_a^l in Theorem 3 satisfies $m_a^l \le \tilde{m}_a^l$, and $\tilde{m}_a^l$ in Lemma 2 is a fixed multiple of $\sqrt{\gamma(\boldsymbol{h}^{(k)})}$, so Theorem 5 gives its rate. For the link cost bounds in Corollary 1 and the TSTT bound in Theorem 4, we observe that $\boldsymbol{a}^{(k)} \to \hat{\boldsymbol{a}}$ and $m_a^l \to 0$. Hence, for all sufficiently large k, the intervals $\mathbb{A}_a^l$ used to compute L_a^l and K_a^l lie in the interval $[\max(0, \hat{a}_l - 1), \hat{a}_l + 1]$, which does not depend on k. Since τ_l' and τ_l'' are continuous, L_a^l and K_a^l remain bounded. Thus, the link cost bounds $L_a^l m_a^l$ in Corollary 1 are $O(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|)$, and the second term $\frac{1}{2}\sum_{l\in\mathbb{L}} K_a^l (m_a^l)^2$ of the TSTT bound in Theorem 4 is $O(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|^2)$. The marginal social path costs q_i are also continuous, so Q_a remains bounded, and the first term $Q_a\sqrt{\theta\gamma(\boldsymbol{h}^{(k)})/2}$ of the TSTT bound is $O(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|)$. Combining the two terms, the TSTT bound in Theorem 4 is $O(\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\|)$. Finally, all of these bounds converge to zero as $\|\mathcal{L}(\boldsymbol{h}^{(k)}) - \boldsymbol{h}^{(k)}\| \to 0$. □

Appendix E. Additional results on the convergence and computational cost of the bounds

We now focus on the convergence of the bounds. Specifically, we want to test Theorem 5 and Corollary 3 numerically. Combined with $\|\mathcal{L}(\boldsymbol{h}) - \boldsymbol{h}\| \le (1 + L_{\mathcal{L}})\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$ from Section 5, they imply that the objective bound $\gamma(\boldsymbol{h})$ should converge to zero at least as fast as $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^2$, and that the path flow, link flow, and TSTT bounds should converge to zero at least as fast as $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$. Since these results describe the bounds close to equilibrium, we only consider iterates after they first reach an RGAP of 10^{-4}. In Figure 6, we plot the objective, link flow, and TSTT bounds for the Chicago Sketch network, together with $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^2$ for the objective bound and $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$ for the other two. As we can observe, the objective bound almost coincides with $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^2$, while the link flow and TSTT bounds stay parallel to $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$, indicating that the ratio between each bound and the corresponding power of $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$ does not grow as the iterates approach equilibrium, as expected from Theorem 5 and Corollary 3.

Table 8: Smallest constant c such that a bound is at most $c\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^p$ at every iterate after the current iterate crosses an RGAP of 10^{-4} ($\theta = 0.5$).

Network	**Objective**, $\gamma(\boldsymbol{h})$	**Path flow**, m_h	**Link flow**, $\|(m_a^l)_{l\in\mathbb{L}}\|$	**TSTT**
Sioux Falls	1×10^{3}	2×10^{3}	2×10^{4}	3×10^{6}
Anaheim	1×10^{-1}	1×10^{1}	4×10^{2}	6×10^{2}
Winnipeg	1×10^{1}	5×10^{1}	3×10^{3}	9×10^{3}
Chicago Sketch	1×10^{1}	2×10^{2}	9×10^{3}	8×10^{4}

Note. $p = 2$ for the objective bound and $p = 1$ for the others.

Table 9: Mean computation time per iteration up to RGAP 10^{-8}, in seconds ($\theta = 0.5$).

Computation	**Anaheim**	**Winnipeg**	**Chicago Sketch**
Iteration of the algorithm	9×10^{-3}	3×10^{-2}	5×10^{-1}
Objective, $\gamma(\boldsymbol{h})$	2×10^{-3}	6×10^{-3}	9×10^{-2}
Path flow, m_h	3×10^{-6}	4×10^{-6}	1×10^{-5}
Link flow, m_a^l	5×10^{-4}	2×10^{-3}	2×10^{-3}
Link cost, $L_a^l m_a^l$	4×10^{-4}	1×10^{-3}	1×10^{-3}
TSTT	1×10^{-3}	4×10^{-3}	6×10^{-2}
All bounds	4×10^{-3}	1×10^{-2}	2×10^{-1}
Time in bounds (%)	43	41	33

Note. Iteration of the algorithm: mean time of one `FP-BB1` iteration including the computation of all the bounds. Time in bounds: The share of an iteration with bounds that is spent computing the bounds.

To compare the four networks, Table 8 reports the smallest constant c such that each bound is at most $c\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|^p$ at every iterate after the RGAP first crosses 10^{-4}, where $p = 2$ for the objective bound and $p = 1$ for the other bounds. The constants differ considerably across networks. For instance, for the objective bound, c ranges from 10^{-1} in the Anaheim network to 10^3 in the Sioux Falls network. This is consistent with Equation (91), whose constant depends on the network through $\theta \min_j \hat{h}_j$ and, when $\|\mathcal{L}(\boldsymbol{h}) - \boldsymbol{h}\|$ is replaced by $\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|$, through $L_{\mathcal{L}}$.

Next, we present results on the computational cost of computing each of the bounds. Table 9 reports the mean time of an iteration of the algorithm, including all the bounds, and of each bound for the Anaheim, Winnipeg, and Chicago Sketch networks. Since the bounds build on one another, each bound row shows only the additional time required to compute that bound once the bounds it depends on are available. For instance, the path flow, link flow, link cost, and TSTT bounds all require $\gamma(\boldsymbol{h})$, but the time to compute $\gamma(\boldsymbol{h})$ is included only in the objective row. It can be observed that, in all three networks, computing all the bounds at every iteration takes between a third and 43% of the time of an iteration. Among the individual bounds, $\gamma(\boldsymbol{h})$ and the TSTT bound take the longest.

In most practical settings, however, the bounds would not be computed at every iteration. Either they would be computed once at the end of the procedure, to give an upper bound on the error in the resulting path flows, link flows, link costs, and TSTT, or they would be checked at a ladder of RGAP values, such as 10^{-4}, 10^{-5}, and 10^{-6}, to stop the assignment as soon as the required accuracy in the metric under consideration is achieved. In either case, each check costs less than one iteration, so the bounds add the computation of only a few iterations while they can save the computation of many more iterations.

Finally, all our computations use extended precision (long double) floating-point numbers to reduce rounding errors when comparing the bounds with a reference solution computed to an RGAP of 10^{-12}. In a simple test of the main operations in our implementation (sparse matrix-vector products, exponentials,

and logarithms) on the machine used in our experiments, long doubles were about three to four times slower than doubles.

Appendix F. Tightness of the bounds

In this appendix, we ask whether the bounds could be reduced by a fixed factor for all problems satisfying our assumptions. To show that they cannot, it suffices to construct a family of problems in which the ratio of each error to its bound can be made arbitrarily close to one. To this end, we will use the network in Figure 7, with a single OD pair w of demand $d_w = 1$ served by two parallel links. We number the two paths so that path i consists only of link i, and hence $a_1 = h_1$ and $a_2 = h_2$. Further, $d_{\max} = d_w = 1$. Since each link is used by exactly one of the two paths, $\mathbb{W}_l = \{w\}$ and $S_l = 1$ for both links. We also set $\theta = 1$. We will look at a family of networks indexed by $r \in (0, 1/2)$. We will first compute the flow errors and $\gamma(\boldsymbol{h})$, which appears in every bound. We will then take each bound in turn, compute it together with the corresponding error, and show that their ratio tends to one as $r \to 0$.

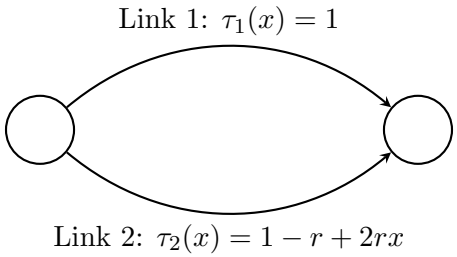


Figure 7: Two-link network used to examine tightness. A single OD pair w with demand $d_w = 1$ is served by two parallel links, each of which forms one path.

Suppose the link costs are $\tau_1(x) = 1$ and $\tau_2(x) = 1 - r + 2rx$, where x denotes the flow on the link. The path flows are the same as the link flows in this example. We observe that the equilibrium is $\hat{h}_1 = \hat{h}_2 = 1/2$ for every r, because at these flows both links cost $\tau_1(1/2) = \tau_2(1/2) = 1$, so the logit formula splits the demand equally between them. We take the current solution to be $h_1 = 1/2 - r$ and $h_2 = 1/2 + r$. The true errors at the current solution are

$$\begin{aligned} &\|\boldsymbol{h} - \hat{\boldsymbol{h}}\| = \sqrt{2}\, r, \qquad \|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1 = 2r, \\ &|a_l - \hat{a}_l| = r, \qquad l \in \mathbb{L}. \end{aligned} \tag{92}$$

We first compute $\gamma(\boldsymbol{h})$. At the current flows, the path costs are $\mathcal{C}_1(\boldsymbol{h}) = 1$ and $\mathcal{C}_2(\boldsymbol{h}) = 1 + 2r^2$, so by Equation (4) the target flows are

$$\begin{aligned} \mathcal{L}_1(\boldsymbol{h}) &= \frac{1}{1 + e^{-2r^2}}, \\ \mathcal{L}_2(\boldsymbol{h}) &= \frac{e^{-2r^2}}{1 + e^{-2r^2}} = 1 - \mathcal{L}_1(\boldsymbol{h}). \end{aligned} \tag{93}$$

To expand these, we use $e^{-2r^2} = 1 - 2r^2 + O(r^4)$, so that $1 + e^{-2r^2} = 2(1 - r^2) + O(r^4)$. Using $1/(1 - r^2) = 1 + r^2 + O(r^4)$ in Equation (93) then gives

$$\begin{aligned} \mathcal{L}_1(\boldsymbol{h}) &= \frac{1}{2}\big(1 + r^2\big) + O(r^4), \\ \mathcal{L}_2(\boldsymbol{h}) &= 1 - \mathcal{L}_1(\boldsymbol{h}) = \frac{1}{2} - \frac{r^2}{2} + O(r^4). \end{aligned} \tag{94}$$

From the definition of ψ_h in Equation (13), $\psi_h(\boldsymbol{h}) = \beta(\boldsymbol{h}) + \eta(\boldsymbol{h})$ and $\psi_h(\mathcal{L}(\boldsymbol{h})) = \beta(\boldsymbol{h}) + \mathcal{C}(\boldsymbol{h})^T\big(\mathcal{L}(\boldsymbol{h}) - \boldsymbol{h}\big) + \eta(\mathcal{L}(\boldsymbol{h}))$. The term $\beta(\boldsymbol{h})$ cancels, so

$$\begin{aligned}\gamma(\boldsymbol{h}) &= \psi_h(\boldsymbol{h}) - \psi_h(\mathcal{L}(\boldsymbol{h})) \\ &= \mathcal{C}(\boldsymbol{h})^T\big(\boldsymbol{h} - \mathcal{L}(\boldsymbol{h})\big) + \eta(\boldsymbol{h}) - \eta(\mathcal{L}(\boldsymbol{h})).\end{aligned} \tag{95}$$

We compute the three terms separately. For the cost term, the current flows and the target flows in Equation (94) give

$$\begin{aligned}h_1 - \mathcal{L}_1(\boldsymbol{h}) &= \left(\frac{1}{2} - r\right) - \left(\frac{1}{2} + \frac{r^2}{2}\right) + O(r^4) \\ &= -r - \frac{r^2}{2} + O(r^4), \\ h_2 - \mathcal{L}_2(\boldsymbol{h}) &= \left(\frac{1}{2} + r\right) - \left(\frac{1}{2} - \frac{r^2}{2}\right) + O(r^4) \\ &= r + \frac{r^2}{2} + O(r^4).\end{aligned} \tag{96}$$

Substituting Equation (96) together with $\mathcal{C}_1(\boldsymbol{h}) = 1$ and $\mathcal{C}_2(\boldsymbol{h}) = 1 + 2r^2$, we obtain

$$\mathcal{C}(\boldsymbol{h})^T\big(\boldsymbol{h} - \mathcal{L}(\boldsymbol{h})\big) = 2r^3 + O(r^4). \tag{97}$$

Since $\theta = 1$, the entropy at the current flows is

$$\begin{aligned}\eta(\boldsymbol{h}) &= h_1 \ln h_1 + h_2 \ln h_2 \\ &= -\ln 2 + 2r^2 + O(r^3).\end{aligned} \tag{98}$$

Similarly, substituting the target flows from Equation (94), the entropy at the target flows is

$$\begin{aligned}\eta(\mathcal{L}(\boldsymbol{h})) &= \mathcal{L}_1(\boldsymbol{h}) \ln \mathcal{L}_1(\boldsymbol{h}) \\ &\quad + \mathcal{L}_2(\boldsymbol{h}) \ln \mathcal{L}_2(\boldsymbol{h}) \\ &= -\ln 2 + O(r^3).\end{aligned} \tag{99}$$

Substituting Equations (97), (98), and (99) into Equation (95) gives

$$\begin{aligned}\gamma(\boldsymbol{h}) &= 2r^3 + \big(-\ln 2 + 2r^2\big) - \big(-\ln 2\big) + O(r^3) \\ &= 2r^2 + O(r^3),\end{aligned} \tag{100}$$

so that $\gamma(\boldsymbol{h})/r^2 \to 2$ as $r \to 0$. We now examine each bound in turn.

Objective bound. Since $\phi = \beta + \eta$, the actual error in the SUE objective is

$$\begin{aligned}\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) &= \big(\beta(\boldsymbol{h}) - \beta(\hat{\boldsymbol{h}})\big) \\ &\quad + \big(\eta(\boldsymbol{h}) - \eta(\hat{\boldsymbol{h}})\big).\end{aligned} \tag{101}$$

For the cost term, the definition of β in Equation (6) gives

$$\begin{aligned}\beta(\boldsymbol{h}) &= \int_0^{h_1} 1\,dx + \int_0^{h_2} (1 - r + 2rx)\,dx \\ &= h_1 + (1 - r)h_2 + rh_2^2.\end{aligned} \tag{102}$$

Substituting the current flows $h_1 = \frac{1}{2} - r$, $h_2 = \frac{1}{2} + r$ and the equilibrium flows $\hat{h}_1 = \hat{h}_2 = \frac{1}{2}$ into Equation (102),

$$\begin{aligned}\beta(\boldsymbol{h}) &= 1 - \frac{r}{4} + r^3, \\ \beta(\hat{\boldsymbol{h}}) &= \frac{1}{2} + \frac{1 - r}{2} + \frac{r}{4} = 1 - \frac{r}{4},\end{aligned} \tag{103}$$

so $\beta(\boldsymbol{h}) - \beta(\hat{\boldsymbol{h}}) = r^3$. For the entropy term, $\eta(\hat{\boldsymbol{h}}) = \frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2} = -\ln 2$, and Equation (98) gives $\eta(\boldsymbol{h}) = -\ln 2 + 2r^2 + O(r^3)$, so $\eta(\boldsymbol{h}) - \eta(\hat{\boldsymbol{h}}) = 2r^2 + O(r^3)$. Substituting both terms into Equation (101) gives

$$\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}}) = r^3 + 2r^2 + O(r^3) = 2r^2 + O(r^3). \tag{104}$$

Since the bound in Theorem 1 is $\gamma(\boldsymbol{h})$, Equations (100) and (104) give the ratio

$$\frac{\phi(\boldsymbol{h}) - \phi(\hat{\boldsymbol{h}})}{\gamma(\boldsymbol{h})} = \frac{2r^2 + O(r^3)}{2r^2 + O(r^3)} \to 1. \tag{105}$$

Path flow bounds. The ℓ_2 bound in Theorem 2 is $m_h = \sqrt{\theta d_{\max}\gamma(\boldsymbol{h})} = \sqrt{\gamma(\boldsymbol{h})}$, and the per-OD ℓ_1 bound is $\sqrt{2\theta d_w \gamma(\boldsymbol{h})} = \sqrt{2\gamma(\boldsymbol{h})}$. Since $\gamma(\boldsymbol{h})/r^2 \to 2$ by Equation (100), the path flow errors in Equation (92) give the two ratios

$$\begin{aligned} \frac{\|\boldsymbol{h} - \hat{\boldsymbol{h}}\|}{m_h} &= \frac{\sqrt{2}\,r}{\sqrt{\gamma(\boldsymbol{h})}} \to 1, \\ \frac{\|\boldsymbol{h}^w - \hat{\boldsymbol{h}}^w\|_1}{\sqrt{2\gamma(\boldsymbol{h})}} &= \frac{\sqrt{2}\,r}{\sqrt{\gamma(\boldsymbol{h})}} \to 1. \end{aligned} \tag{106}$$

Link flow bound. Since $S_l = 1$, the bound in Lemma 2 is $\tilde{m}_a^l = \sqrt{\theta\gamma(\boldsymbol{h})/2} = \sqrt{\gamma(\boldsymbol{h})/2}$ for both links. Link 1 has a constant cost, so $R_a^1 = 0$ and the bound in Theorem 3 is $m_a^1 = \tilde{m}_a^1 = \sqrt{\gamma(\boldsymbol{h})/2}$. For link 2, $\tau_2'(x) = 2r$, so $R_a^2 = 2r$ and $m_a^2 = \tilde{m}_a^2/\sqrt{1 + r/2} = \sqrt{\gamma(\boldsymbol{h})/(2+r)}$. With the link flow errors in Equation (92), the ratios satisfy

$$\frac{|a_1 - \hat{a}_1|}{m_a^1} = \frac{\sqrt{2}\,r}{\sqrt{\gamma(\boldsymbol{h})}} \to 1, \qquad \frac{|a_2 - \hat{a}_2|}{m_a^2} = \frac{\sqrt{2+r}\,r}{\sqrt{\gamma(\boldsymbol{h})}} \to 1. \tag{107}$$

Link cost bound. The cost of link 1 is constant, so both its error and its bound are zero. For link 2, Equation (92) gives the actual error $|\tau_2(a_2) - \tau_2(\hat{a}_2)| = 2r|a_2 - \hat{a}_2| = 2r^2$. Since $\tau_2'(x) = 2r$, Corollary 1 gives $L_a^2 = 2r$. Using $m_a^2 = \sqrt{\gamma(\boldsymbol{h})/(2+r)}$ from the link flow bound, the ratio satisfies

$$\begin{aligned} \frac{|\tau_2(a_2) - \tau_2(\hat{a}_2)|}{L_a^2 m_a^2} &= \frac{2r^2}{2r\sqrt{\gamma(\boldsymbol{h})/(2+r)}} \\ &= \frac{\sqrt{2+r}\,r}{\sqrt{\gamma(\boldsymbol{h})}} \to 1. \end{aligned} \tag{108}$$

TSTT bound. The TSTT is $\zeta(\boldsymbol{a}) = a_1 + (1-r)a_2 + 2ra_2^2$, which equals 1 at equilibrium and $1 + r^2 + 2r^3$ at the current flows. The actual error is therefore $\zeta(\boldsymbol{a}) - \zeta(\hat{\boldsymbol{a}}) = r^2 + 2r^3$. For the bound in Theorem 4, the marginal social path costs are $q_1 = 1$ and $q_2 = \tau_2(a_2) + a_2\tau_2'(a_2) = 1 + r + 4r^2$, so $Q_a = \sqrt{d_w}\,(q_2 - q_1) = r + 4r^2$. Moreover, since the cost of link 1 is constant and the cost of link 2 is linear, the second derivative of ζ with respect to each link flow does not depend on the flow, and equals 0 for link 1 and $4r$ for link 2. Its maximum over $\mathbb{A}_a^l$ therefore gives $K_a^1 = 0$ and $K_a^2 = 4r$. Using $(m_a^2)^2 = \gamma(\boldsymbol{h})/(2+r)$, the bound is therefore

$$\begin{aligned} &Q_a\sqrt{\frac{\theta\gamma(\boldsymbol{h})}{2}} + \frac{1}{2}K_a^2(m_a^2)^2 \\ &\quad = (r + 4r^2)\sqrt{\frac{\gamma(\boldsymbol{h})}{2}} + \frac{2r\gamma(\boldsymbol{h})}{2+r}. \end{aligned} \tag{109}$$

Dividing both the error and the bound in Equation (109) by r^2, and using $\gamma(\boldsymbol{h})/r^2 \to 2$ from Equation (100), which also gives $\gamma(\boldsymbol{h})/r \to 0$, the ratio satisfies

$$\begin{aligned} &\frac{\zeta(\boldsymbol{a}) - \zeta(\hat{\boldsymbol{a}})}{(r + 4r^2)\sqrt{\gamma(\boldsymbol{h})/2} + 2r\gamma(\boldsymbol{h})/(2 + r)} \\ &\quad = \frac{1 + 2r}{(1 + 4r)\sqrt{\gamma(\boldsymbol{h})/(2r^2)} + 2\gamma(\boldsymbol{h})/\big((2 + r)r\big)} \to 1. \end{aligned} \tag{110}$$

Since every ratio tends to one, none of the bounds examined above can be multiplied by a fixed factor smaller than one uniformly over the class of problems we consider.